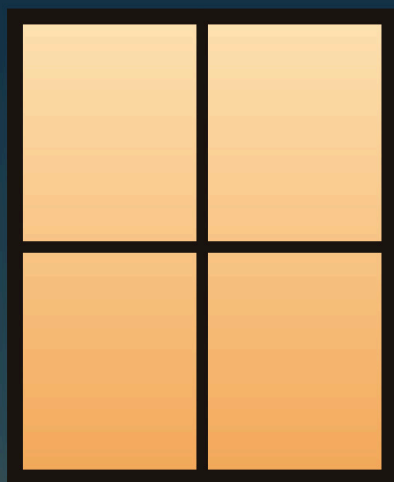


TIME · DIGNITY · THE CARE STILL WORTH BUILDING

WHAT TIME IS FOR



*A Human Account of Kidney Disease, Waiting,
and the Care Still Worth Building*

1,092 HOURS A YEAR · ONE QUESTION

JEFF PARKE

FOUNDER · COLD ISCHEMIA FOUNDATION

WHAT TIME IS FOR

*A Human Account of Kidney Disease, Waiting, and
the Care Still Worth Building*

JEFF PARKE

COLD ISCHEMIA FOUNDATION

What Time Is For

A Human Account of Kidney Disease, Waiting, and the Care Still Worth Building

Copyright © 2026 Jeff Parke. All rights reserved.

Published by the Cold Ischemia Foundation, Ellenton, Florida. coldischemia.foundation

First edition, October 2026.

No part of this book may be reproduced, stored in a retrieval system, or transmitted in any form or by any means without the prior written permission of the publisher, except for brief quotations used in reviews or scholarly work with full attribution.

Educational purpose. This book describes systems and people. It is not medical, legal, or financial advice and does not replace the counsel of your own clinicians. In a medical emergency call 911. If you or someone you love is in emotional crisis, call or text 988.

On people and scenes. Scenes that open chapters are composites built from documented patterns. No individual is described unless a source is cited.

On accuracy. Statements about organizations, investigations, and legal matters describe the sources cited. Sources marked with a dagger (†) have bibliographic details that should be completed from the original before any future edition. Corrections are welcome through the Foundation's contact page.

Quotations. The epigraphs are drawn from works in the public domain.

*For the ones who sit with the machine, and
for the ones who sit with them.*

*“To every thing there is a season, and a time to every
purpose under the heaven.”*

— Ecclesiastes 3:1 (King James Version, 1611)

Contents

Prologue: Four Hours	7
A Note Before We Begin	11
PART ONE: THE HOURS	13
Chapter 1: The Half-Time Job	13
Chapter 2: The Hour After	18
Chapter 3: The Work Question	23
Chapter 4: The Little Platoon	28
Chapter 5: The Quiet Option	33
PART TWO: THE DOORS	38
Chapter 6: The Disease No One Feels	38
Chapter 7: The Navigator	43
Chapter 8: Dialysis at the Kitchen Table	48
Chapter 9: The Distance Between Here and Care	53
Chapter 10: The Storm	59
Chapter 11: The Emergency Chair	63
PART THREE: THE GIFT	67
Chapter 12: The Long Chain	67
Chapter 13: The Receipt	73
Chapter 14: The Hallway	77
PART FOUR: THE MACHINERY OF HOPE	82
Chapter 15: Three in the Morning	82
Chapter 16: The Odd Machines	88
Chapter 17: A Kidney Is Not Forever	94
Chapter 18: The Handoff	99
Chapter 19: The Pharmacy Counter	103
Chapter 20: Who Is Keeping Count	107
Chapter 21: What a Conservative Would Keep	113
Epilogue: A Tuesday Worth Wanting	118
Appendix A: Source Notes	121
Appendix B: References	130
About the Author	138

Prologue: Four Hours

“I have discovered that all the unhappiness of men arises from one single fact, that they cannot stay quietly in their own chamber.”

Blaise Pascal, *Pensées* (Trotter translation, 1908)

Pascal believed the trouble with people was that they could not sit still. He never met anyone whose treatment required it.

Picture a room with a row of recliners and a window nobody looks out of. The chairs are vinyl, the color of weak tea. Each one faces a machine the size of a bar refrigerator, and each machine hums at a slightly different pitch, so the room has a kind of chord in it. A television is bolted high in a corner, tuned to a program no one chose. It is one o'clock on a Tuesday. Nobody here is sick in the way the movies show sickness. They are reading, dozing, doing crosswords, answering email, watching their own blood travel in a loop through clear tubing and come back clean. They are, for the next four hours, exactly where they are supposed to be.

That scene is a composite, built from what patients, nurses, and researchers have described. No single person stands in it. I begin with it because I have come to believe that the most important fact about

kidney failure is also the one least often counted. It is not the number of people waiting for an organ, though that number is large. It is not the cost, though the cost is larger. It is the hours.

Let me do the arithmetic, slowly, because slowness is the point. A common schedule for in-center hemodialysis is three sessions a week of about four hours each. Over fifty-two weeks that is 156 sessions and 624 hours in the chair. Now add what comes afterward. In an online survey of 359 patients who dialyze in centers, the median time to recover from a session was three hours, and four in ten said it took four hours or longer. The most common complaint was feeling washed out, reported by 62 percent, followed by cramps at 44 percent and symptoms of low blood pressure at 42 percent (Source Notes S1). Three hours of recovery across 156 sessions is another 468 hours.

Together, 1,092 hours. A full-time job occupies about 2,080 hours in a year. By this reckoning, kidney failure treated in a center takes something like half of a full-time job, and the person holding it is not paid, cannot quit, and is often too tired to do any other job at all. I need to be honest about the limits of that figure. The schedule is typical and not universal, the recovery number comes from one modest survey, and recovery time overlaps with sleep and rest in ways no survey can untangle. Travel to and from the clinic is not counted at all. The point is not that the sum is exact. The point is that it is the right kind of sum, and that almost nobody does it.

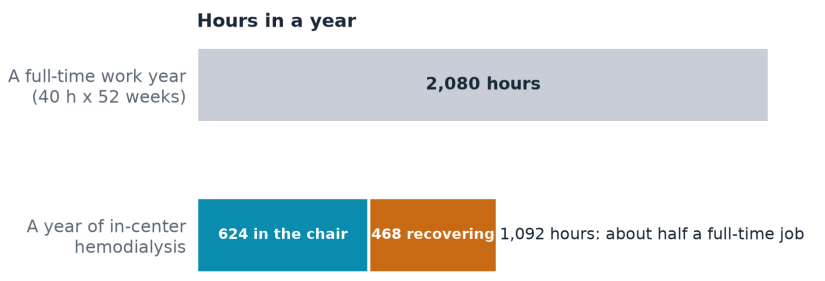


Figure 0.1. A year of in-center hemodialysis in hours, compared with a full-time work year. Illustrative arithmetic from a common three-session schedule and a published median recovery time; travel excluded.

I have lived long enough inside this world, in more than one role, to know the sound that rooms like this make, and the sound that a family makes when it is doing a sum like this one in the parking lot. I will not tell you my own story. It belongs to me and to the few people who carried me through it. What I can offer instead is a way of looking, which is to ask of every part of the American kidney ecosystem a question so plain it sounds naive. What is this doing to people's time?

The question has an old pedigree. The psalmist asked God to teach us to number our days. The Stoics made an entire discipline of the observation that life is not short so much as spent carelessly. In our era the question has been reduced to productivity, which is a shrunken version of it. But in a dialysis chair, the old question returns at full size. If you were given a certain number of afternoons, what would you want them for? And what do the rest of us owe a person whose afternoons we have, by accident of biology, quietly claimed?

This book is my attempt to take that question seriously, in two registers at once. One is existential, in the plain sense of the word: it concerns finitude, choice, and the strange dignity of deciding how to spend a limited life. The other is conservative, in the older sense of the

word: it concerns what is worth preserving, the humility to ask why a thing stands before tearing it down, and a stubborn faith in the small institutions, a family, a congregation, a town's clinic, a hospital's coordinator, that do most of the real work of caring while the grand systems get the credit.

You will find that the chapters do not follow one another like a case being made. They are more like rooms in a house, each with a different window. One looks onto the dialysis afternoon. One looks onto the quiet decision some people make to decline dialysis altogether. One looks onto the person who drives, the person who gives, the family in a hallway, the surgeon's telephone at three in the morning, the machines that may someday set the afternoons free. I have tried to bring current evidence into every room, drawn from the last year wherever it exists, and to say plainly where the evidence ends.

I will make you one promise and one request. The promise is that I will not use a statistic to frighten you when a sentence will do, and I will not use a story to persuade you of something the data do not support. The request is that you read slowly. The people in these chairs have been taught patience whether they wanted the lesson or not. We might at least match it.

A Note Before We Begin

“Felix qui potuit rerum cognoscere causas.”

Virgil, Georgics II.490 (“Fortunate is the one who has been able to
learn the causes of things.”)

A few plain words about how this book was made, because a book that asks you to trust it owes you the details.

The scenes are composites. When a chapter opens with a picture of a person or a room, it is built from documented patterns, not from an individual. I have given no one a real name, and I have invented no statistics inside any scene. Whenever I describe an actual person or event, I cite the source.

The facts are graded. Every claim carries an inline citation. Behind each one is a rating I assigned in the source notes at the back. A means a primary document, a registry, or a peer-reviewed paper whose key figures I could confirm. B means a credible secondary account. C means a figure from a single source or from a party with a stake in the result, such as a manufacturer or an advocacy campaign. Material graded C is always labeled as preliminary where it appears.

Where I could not see the whole page. The research tools I used returned summaries and bibliographic details, not the full text of every article or agency page. A dagger (†) after a citation means the central figures were confirmed but some detail, such as a complete author list, should be completed from the original before any future edition. I would rather show you that gap than hide it.

The date. This was written in October 2026. I have leaned on developments from the preceding six months wherever they exist, and I say so when the newest figure is older. A few landmark studies appear as background and are labeled as such.

Disclosure. The Cold Ischemia Foundation accepts no donations, grants, or sponsorships, and charges posted fees only for services that individuals request. The Foundation publishes educational material about oxygen-carrier technology, including the work of a company called BHOC Therapeutics. The author reports no financial interest in that company or in any other organization discussed here. The Foundation's own position is that the technology is a research concept whose benefits are unproven, and this book repeats that position without softening it.

Not medical or legal advice. This book describes systems and people. It is not clinical guidance. In a medical emergency, call 911. If you or someone you love is in emotional crisis, call or text 988.

PART ONE
THE HOURS

CHAPTER 1

The Half-Time Job

*“So teach us to number our days, that we may apply our hearts
unto wisdom.”*

Psalm 90:12 (King James Version, 1611)

Picture a woman I will call Ruth. She does not exist, and her afternoons are real. She retired from bookkeeping a few years ago, and she is exceptional at one thing, which is arriving. Her ride comes at 12:15. By 12:40 she is weighed, wrapped in a warm blanket she brought from home, and plugged into the chair. By 1:00 the machine has begun to do what her kidneys no longer can. She has brought a paperback and a thermos. She will read perhaps forty pages. She knows the names of the nurses’ children. She knows who sleeps, who talks, and who gets quiet around hour three.

At 5:15 she is unhooked. She will not go to dinner with her sister, because she will not feel like it. She will sit in the passenger seat and watch the town go by in a way that is a little like watching a film of herself, and then she will be home, and she will be tired in a bone-deep, bargain-basement way that rest does not quite fix. On Wednesday she will feel better. On Thursday she will start counting backward toward Friday's chair.

If you asked Ruth to rank what matters most about her treatment, you might expect her to say staying alive. She would, of course, say that. But researchers have asked patients, and the answer gets more interesting after the obvious.

What people asked for

An international project called SONG-HD set out to discover which outcomes ought to be measured in trials of hemodialysis. It did not simply ask experts. It surveyed patients and caregivers alongside clinicians, in three rounds, across 73 countries. More than 1,100 people took part in the first round, including 202 patients and caregivers and 979 health professionals, and 838 stayed through the third (Developing a set of core outcomes for trials in hemodialysis, 2017†).

The results were quietly startling. Where the two groups disagreed, they disagreed in a pattern. Patients and caregivers rated ability to travel higher than clinicians did, by nearly a full point on the scale. They rated time free from dialysis higher, by half a point. They gave higher marks as well to the adequacy of dialysis and, perhaps most tenderly, to a symptom the clinical literature barely names: feeling washed out after a session. Clinicians, for their part, rated

mortality higher, along with hospitalization, drops in blood pressure, complications of vascular access, depression, and cardiovascular disease.

Neither list is wrong. A doctor is right to worry about death, and a patient is right to want to see her grandchildren in another state. But look at what the patients’ list has in common. Travel, free time, feeling like oneself in the evening: these are not measures of the body’s chemistry. They are measures of a life that has room in it. I should say that I could not confirm from the summaries I could reach which outcomes made the project’s final core set, and I would not want to put words in its authors’ mouths. The shape of the disagreement is enough. Those who live with the machine and those who tend it agree that it matters, and differ on what it is for.

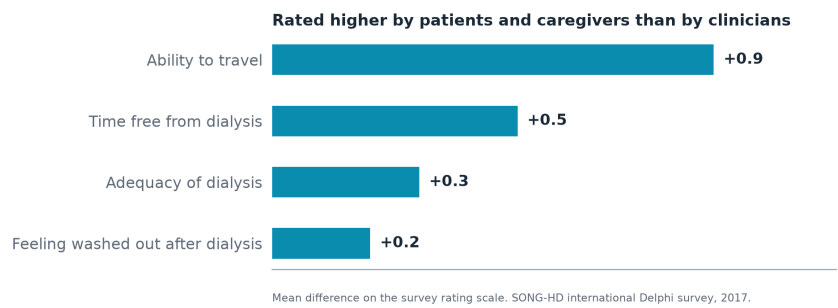


Figure 1.1. Outcomes that patients and caregivers rated higher than clinicians did, as mean differences on the study’s rating scale. Source: SONG-HD international Delphi survey (2017†).

What it does to the mind

Hours do not only pass. They leave residue. A 2025 meta-analysis pooled 137 studies covering 21,608 hemodialysis patients and estimated that 45.8 percent of them had depression, with a confidence interval of 42.0 to 49.6 percent. Prevalence was lowest in

high-income countries, at 38.9 percent, which is still more than one in three (Prevalence of depression in hemodialysis patients, 2025†). The studies differ in method, and they were mostly cross-sectional snapshots, so no one should read the figure as a precise census. But the authors' conclusion is hard to argue with: screening and identifying those at risk is valuable, and I found no sign that it is routine.

Loneliness belongs in the same conversation. A large population-based cohort of people with chronic kidney disease, not dialysis patients specifically, found that those in the most socially isolated group had a 33 percent higher risk of death, and that isolation was associated with a loss of about two years of life expectancy for women and nearly three for men. Those who reported loneliness had a 10 percent higher risk (Association of social isolation and loneliness with the incidence and life expectancy of chronic kidney disease, 2026†). Reviews of the qualitative literature add the texture: the schedule of treatment dominates the calendar, which makes it hard to plan a holiday or a family dinner, and social support appears to blunt the effect.

Here a conservative sensibility has something to say that a purely clinical one might miss. We tend to treat loneliness as a feeling and dialysis as a procedure, and to place them in different departments. But they are one thing in one body in one chair. A person whose treatment consumes the afternoons that friendship usually occupies is not being treated for kidney failure alone. She is being treated for the loss of her ordinary life, and the ordinary life is where the little platoons live, the neighbors, the congregation, the card game. The medicine that ignores them is incomplete in a way no lab value reveals.

The question the chair asks

I do not mean to suggest that anything here can be solved by kindness. Machines are necessary, and the nurses who run them are, in my experience, among the most patient people in medicine. I mean something narrower. A system reveals its priorities by what it counts. Ours counts deaths, hospitalizations, and transplant volumes with admirable precision. It counts Ruth's Tuesday not at all.

So let me ask the question the chair asks, and let me ask it of every chapter ahead. If this person had a thousand and ninety-two hours back, what would she do with them? Whatever the answer, it is the measure of what we are really building toward. A better machine, a quicker transplant, a quieter way of living without either: each is a different answer to the same plea, which is simply to be given back the afternoon.

CHAPTER 2

The Hour After

“It is vain for you to rise up early, to sit up late, to eat the bread of sorrows: for so he giveth his beloved sleep.”

Psalm 127:2 (King James Version, 1611)

Picture the first hour after a dialysis session, which is the hour no one writes down. Ruth is in the passenger seat with a warm blanket over her knees and a small paper cup of juice she did not want. The nurse has pressed gauze to her arm and asked whether she is all right, and she has said yes, because that is the answer that gets you out the door. The light outside is the color of an old photograph. She feels, as she always does, like a cloth that has been wrung out and hung up wet.

She will not describe it that way to her doctor. Doctors ask about blood pressure, about the numbers on the printout, about whether she is “tolerating treatment,” and she is, in the narrow sense. She is not being rushed to a hospital. She is not cramping so badly she needs the session cut short. She is merely a person who has spent four hours having her blood cleaned and has been returned to her own life with the batteries drained.

Patients have a word for this and clinicians, until recently, mostly did not. They say they are washed out.

A symptom with a clock

When researchers finally asked how long the hour after lasts, the answers were longer than anyone had put on a chart. In the online survey of 359 in-center patients described in the Prologue, the median time to recover from a session was three hours, with a range from none to a full day, and 40 percent of patients said it took four hours or longer. Fatigue, or feeling washed out, was the most common symptom at 62 percent, followed by cramps at 44 percent and the signs of low blood pressure at 42 percent (Source Notes S1). A general review puts the typical median at two to four hours and says about a quarter of patients report recovery taking more than six (Source Notes S2).

What interests me is how little that figure resembles any measure the system tracks. A recovery time of three hours is not reported on a quality dashboard. It is not required by a payment model. It does not appear in the federal data. Yet the patients who took part in the international survey described in Chapter 1 put exactly this experience, feeling washed out after dialysis, on the list of things that mattered more to them than they did to the professionals.

What shortens it

If the hour after is a real harm, the obvious question is whether anyone has found a way to shorten it. The evidence is thinner than the question deserves, but it is not empty.

A French cohort study of 536 patients found that 15 percent said the washed-out feeling lasted more than twelve hours. Of all the features of the dialysis prescription the researchers examined, only two were associated with longer recovery: a bigger drop in the blood's sodium during the session, and a lower frequency of treatment, two sessions a week against three or more (Source Notes S2). It is an observational finding and cannot prove that adding a session would shorten the recovery, but it points in a direction.

Another lever is the temperature of the dialysis fluid. In a 2024 quasi-experimental study of 92 patients in Pakistan, fluid at 36 degrees Celsius was associated with less frequent and shorter post-dialysis fatigue than fluid at 37 degrees, without a difference in low blood pressure (Source Notes S3). An older non-randomized crossover study of 50 patients compared 35 and 37 degrees, and 76 percent reported feeling more energetic with the cooler fluid (Source Notes S3). A study that changed patients' prescriptions reported that median recovery time fell from 210 minutes to 60 minutes after six months and 105 minutes after twelve (Source Notes S4). Registered trials of cooler fluid exist, including one in Iran with 46 patients and one in Turkey, and I found no published results from either.

The best-known trials of more frequent dialysis asked different questions. The FHN Daily Trial randomized 245 patients to dialysis six times a week or three, and found that frequent dialysis significantly reduced the mass of the heart's left ventricle by 13.1 grams, with an interval of 5.0 to 21.3 (Source Notes S5). The companion nocturnal trial showed similar trends that did not reach statistical significance, though patients on nocturnal dialysis had better control of blood phosphate and blood pressure. Neither trial, as far as I could determine, made recovery time or fatigue its main outcome.

The gap

That last observation is the point of this chapter. I could not find a randomized trial in which the length of the hour after, or the fatigue that fills it, was the primary outcome of changing the schedule. The things that have been tested are the things that fit the traditional vocabulary of nephrology: heart mass, phosphate, blood pressure, survival. The thing that patients named, in the rare moments they were asked, has hardly been tested at all.

I do not think this is an accident or a scandal. It is an ordinary consequence of what a field decides to count. Trials chase outcomes that are easy to measure and that regulators accept, and recovery time is neither. But it means that a person like Ruth, wondering whether a cooler bath or a longer session or a home machine might give her back her evenings, has almost no evidence to consult. She is being asked to choose in the dark about the one outcome she cares about most.

A vital sign for the evening

There is a small, almost embarrassingly simple remedy. At every visit, a nurse or doctor could ask a patient the question the survey asked: how long does it take you to feel like yourself after treatment? And then write down the answer, in hours, in the same place as the blood pressure. A recovery time recorded each month would reveal at once whether a change in prescription, a different fluid temperature, or a move to home treatment made any difference. It would turn an invisible symptom into a variable, which is the first step toward doing something about it.

This is, in the language of the older conservative tradition, an argument from attention. The small, unglamorous things that make daily life bearable are exactly the things that grand systems overlook, and exactly what wise institutions take care to notice. The hour after is the hour in which a person returns from the machine to the world. A system that wished to treat her as a whole person would want to know how that return goes.

CHAPTER 3

The Work Question

“There is nothing better, than that a man should rejoice in his own works; for that is his portion.”

Ecclesiastes 3:22 (King James Version, 1611)

Picture a man of forty-four who has taught high-school shop for nineteen years. He is good at it. He knows which fifteen-year-old can be trusted with the table saw and which needs another week with the hand plane. Two years ago his kidneys failed, and his treatment, in the plain arithmetic of Chapter 1, now occupies a schedule that does not bend to the bell. He has used up his sick days. His principal is kind and the district is not. At a dinner party, a stranger asks the question Americans ask each other instead of “who are you,” which is “what do you do?” He opens his mouth, and for the first time in his life he does not know what to say.

The man is invented. The question he cannot answer is one the kidney literature has measured with some care.

The numbers on work

A large Danish registry study followed adults between 18 and 65 who began dialysis or received a transplant between 2005 and 2019, linking each person's treatment records to social-benefit records. At the time dialysis began, 11 percent were employed, against 61 percent of matched people from the general population. Among transplant recipients the employment rate was 20 percent, against 67 percent for their matched counterparts. The authors concluded that transplantation leads to only a small increase in employment (Source Notes S6).

Other studies find somewhat higher figures, because they count different things. A systematic review and meta-analysis that pooled studies from 1966 to 2020 reported an employment rate of 26.3 percent among dialysis patients, and rates of 36.9 percent before transplant and 38.2 percent after (Source Notes S7). A United States cohort of people starting dialysis between 1996 and 2013 found 23 to 24 percent working (Source Notes S7). A small 2025 study of 94 transplant recipients in a region of Poland found that 66 percent had returned to work and that 41.9 percent had done so within six months, with return associated with higher education, better finances, and more time since the transplant (Source Notes S8).

The range from 11 percent to 66 percent is wide enough to need explaining, and the explanation is that each study asks a slightly different question. The Danish study measures who has a job at fixed moments, compared with matched people. The Polish study asks whether a recipient ever went back. A meta-analysis blends eras and countries. If I had to carry one number out of this literature, it would

be the Danish one, because it has a comparison group, and it says that kidney failure removes most of a person's working life and that a transplant returns only part of it.

Why work matters beyond money

I will not pretend the economics are small. A person who stops working loses income, employer-sponsored insurance, and the contributions that fund a later life. But the conservative tradition has always held, rightly, that work is also a form of belonging. To have a job is to be needed by someone you are not related to. It is to have a place that would notice your absence. It supplies a structure to the week, and a reason to put on a clean shirt, and a way of being useful that does not depend on anyone's pity.

Chapter 1 described dialysis as a half-time job nobody pays for. This chapter asks what happens when it collides with the job somebody does. An hour of recovery that lasts three, an appointment that cannot move, a body that gives out at the moment the afternoon shift begins: each is a small incompatibility, and together they push people out of work not through any dramatic decision but through the slow arithmetic of a calendar that cannot hold both.

What the literature does not say

I want to be careful about what I can claim. Nothing in the studies I found proves that a particular kind of dialysis, such as treatment at night or at home, allows more people to keep working. It is natural to suppose so, and a person who dialyzes while asleep has more of the day. But I did not find a study that tested it, and I would be reasoning from the structure of the schedule and not from evidence. What the

studies do show is that return to work after transplant is incomplete, and that it varies with education, finances, and time since the operation, which suggests that the barriers are as much social as medical.

What a good employer would ask

There are small, practical gestures that cost an employer very little. Ask what the treatment schedule is, and move the meeting. Let a person leave at noon on the three days that matter and make up the hours at home. Do not require a letter for every appointment. Treat the occasional bad Thursday as a bad Thursday and not as a character flaw. Several of these gestures are matters of ordinary decency that no law needs to compel, and the small institutions in which most people work, the family business, the school, the congregation's office, are exactly the places where they are most easily offered.

The Polish study I mentioned contains one more observation worth carrying forward: return to work was associated with higher education and good financial standing, which means that the people least likely to go back are those with the fewest resources to begin with. A transplant is a medical event, and the return to ordinary life after it is a social one, shaped by whether a person has savings, a skill that can be done part-time, and an employer who can wait. That suggests a place for a modest form of help that does not resemble medicine at all: a vocational conversation early in the course of kidney failure, before the job has been lost and the skill has gone stale.

The question beneath the question

The man in the shop classroom is not asking for charity. He is asking, in effect, whether the people who run institutions can imagine a person like him continuing to belong. A school that offered him a lighter schedule, a hospital that offered dialysis at an hour that fit, an insurer that did not punish him for part-time work, a transplant program that asked in its first interview what the patient did for a living and what he feared losing: each would be an expression of the same basic respect.

The question “what do you do?” has an existential edge that I do not think we should smooth over. For many people, what they do is a large part of who they are, and the loss of it is a bereavement that goes unmarked because nobody died. A medical system that treats the kidneys and ignores the identity has treated half of the patient. Ecclesiastes says that there is nothing better than that a man should rejoice in his own works, for that is his portion. A portion is what is allotted. The least a decent system can do is leave a person’s portion where he can reach it.

CHAPTER 4

The Little Platoon

“...the growing good of the world is partly dependent on unhistoric acts; and that things are not so ill with you and me as they might have been, is half owing to the number who lived faithfully a hidden life.”

George Eliot, *Middlemarch* (1872)

Somebody drives Ruth. We rarely ask who.

In the vocabulary of the field he is a care partner, which is the sort of phrase that sounds like it was invented by a committee and, for once, earned its place. He might be her husband. He might be a son who moved back into his old bedroom, or a neighbor, or a niece. Whoever he is, he keeps the calendar, remembers the medication names, watches for the swelling in her ankles, sits through the appointments, and does the thing nobody else will do, which is to say “I’m worried about her” to a doctor with eleven minutes.

Edmund Burke wrote, in 1790, that “to be attached to the subdivision, to love the little platoon we belong to in society, is the first principle (the germ as it were) of public affections.” (Burke, 1790) He was not writing about dialysis. But I know no better description of

the unit in which kidney care actually takes place. It is not a hospital. It is not a payer. It is two or three people at a kitchen table with a pill organizer.

What the platoon pays

The research on caregivers of dialysis patients is mostly small, from many countries, and cross-sectional, so it shows association and not cause. It is consistent enough to take seriously. Across studies in several settings, the hours a person spends caring are linked to depression, anxiety, and strain, and the burden tends to rise with the length of the caregiving (Source Notes S9). I will not offer a single number for how many hours a week the average care partner gives, because the figures I found were reported in incompatible units and I do not trust them. I would ask instead that you notice the absence of the number. An enterprise that costs the public billions and rests on the unpaid labor of the patient's family has never, so far as I could find, made counting that labor a priority.

The gift of asking

There is one place where the platoon has been put to deliberate use, and the results are instructive. Many people who need a living-donor kidney simply cannot bring themselves to ask. The request is intimate and awkward, and the person most in need of courage is the person with the least to spare. A program at Johns Hopkins, called the Live Donor Champion program, answered with a change of roles. The patient chooses a friend, relative, or community member to speak on her behalf, and the two attend six monthly sessions of about an hour each, learning how to talk about living donation and how to map the networks they belong to.

In an early pilot, 15 candidates with no prospective donor went through the program and were compared with 15 matched candidates who did not. Twenty-five potential donors contacted the center on behalf of the program group. Four participants went on to receive a living-donor transplant and three more had donors in evaluation. Among the matched controls the number was zero (Source Notes S10). A larger follow-up of 163 candidates reported 81 donor referrals, and participants had about five and a half times the odds of at least one referral compared with matched controls on the waiting list (Source Notes S10). That result was presented as a conference abstract and it is observational, and I found no randomized trial of the program. I treat it as promising, not proven.

What moves me is not the statistic but the design. The program did not hire a bigger institution. It trained a friend. It recognized that the strongest asset a patient has is the web of people who already love her, and that the web needs words, permission, and a little coaching to become useful.

A faith-based organization in Israel has taken the same insight a step further. Matnat Chaim, a nonprofit that arranges mostly altruistic donations between strangers, facilitated 494 living kidney donations from its founding in 2009 through the end of 2017, according to a retrospective review (Emergence of an Israel faith-based community organization facilitating live donor kidney transplantation, 2018†). A retrospective count shows scale and not cause, and nobody should read it as proof that a congregation can replace a hospital. It does suggest that a community with a shared moral language can do something a bureaucracy cannot, which is to make generosity feel ordinary.

An institution built like a platoon

The most striking large-scale illustration of this principle may be Spain. In 2025 the country recorded 51.9 deceased donors per million people, one of the highest rates in the world, with 2,547 deceased donors and 408 living donors that year, a little below its own record of 52.6 per million set in 2024 (Ministerio de Sanidad, 2026; La Moncloa, 2025). I will not compare that rate with the American one, because definitions of a donor differ and the comparison invites more heat than light. What interests me is the structure. The Spanish system is organized in three tiers, national, regional, and hospital, and at the base of the pyramid sits a hospital coordinator, usually an anesthesiologist or intensive-care physician working part-time in the role, embedded in the hospital where donation happens (The Spanish model, 2025). The coordinator knows the unit, the staff, and often the family. This is an institution designed on the scale of a platoon.

Whether it can be transplanted to another country is unclear, and its admirers disagree about which feature does the work. But the lesson for a conservative reader seems plain. The nearer a decision sits to the people it affects, the better it tends to go. We have built a national apparatus of extraordinary technical power, and I suspect it would work better if more of its human contact were placed in the hands of people who can be recognized in a hallway.

What we owe the platoon

If the little platoon is where care happens, then policy has a simple obligation, which is to stop treating it as free. That does not necessarily mean wages. It could mean respite, training, a named person to call, a plain-language map of what is coming, or a clinic that

asks the care partner how she is. It would be a modest revolution in a field that has built an elaborate machinery of support for the patient and almost none for the person who stands behind her.

Eliot ended her great novel by honoring the hidden lives that keep the world from being worse than it is. That is the care partner, to the letter. The least we owe such people is to count them.

CHAPTER 5

The Quiet Option

“That to philosophize is to learn to die.”

Michel de Montaigne, essay title, Florio translation (1603)

Picture a woman of eighty-two, whom I will not name, sitting across from a nephrologist whose kindness is obvious and whose clock is ticking. Her kidney function has fallen to the point where dialysis is the next step. The doctor has laid out the plan with a good deal of skill: access surgery, a schedule, a chair. And the woman, who spent her working life as a seamstress and has outlived her husband and one of her children, asks a question that is not on the form. “What would it cost me,” she says, “in afternoons?”

This scene is a composite, and I begin with it because the question she asks is one that American kidney care has barely learned to hear.

An option almost nobody is offered

Conservative kidney management, sometimes called comprehensive conservative care, means choosing to treat kidney failure without dialysis or a transplant, with attention to symptoms, comfort, and the

patient's priorities for the time that remains. It is not "doing nothing." It involves medication, diet, support, and often palliative care. It is also, in the United States, strikingly rare.

In a 2023 analysis of a large cohort of people with chronic kidney disease, only 7 percent of 1,018 participants were planning for conservative management. Among 407 who completed a survey, 18 percent reported ever having a conversation with a provider about forgoing dialysis. The providers, for their part, mostly said they felt comfortable raising the subject, and yet none worked in a setting with a dedicated conservative-management clinic or protocol (Source Notes S11). A qualitative study of twenty nephrologists at Veterans Affairs facilities found that national guidelines call for discussing conservative management with patients at advanced stages, that this rarely happens, and that clinicians feel real emotional strain when they care for older patients who are moving toward dialysis (Source Notes S12).

The reasons are not hard to guess. Dialysis is reimbursed, scheduled, and accompanied by an infrastructure. Conversation is none of those. And physicians trained to rescue find it hard to say that rescue may not be what this person wants.

What the evidence suggests, and what it cannot

Observational studies of adults aged seventy-five and older who have significant illness or functional limitation suggest that survival and quality of life may not differ appreciably between those who start dialysis and those who choose a conservative path. Dialysis is associated with more hospital stays and a smaller chance of dying at home or in hospice. In studies of people approaching kidney failure, a

substantial share said they would give up meaningful life expectancy to reduce the burden of treatment (Source Notes S13). A 2026 qualitative study in JAMA Network Open, drawing on two American randomized trials conducted between 2019 and 2024, examined why older adults choose this path. I could read only its abstract, which describes the approach as infrequently used despite the weight dialysis places on patients and caregivers (Older patients and conservative kidney management, 2026†).

Let me be careful here, because this is the most delicate subject in the book. Observational comparisons of this kind are vulnerable to bias, since the people who choose conservative care are, on average, frailer to begin with. I am not arguing that anyone should forgo dialysis, and I would be appalled if this chapter were read as an argument for fewer starts. I am arguing for something simpler and more demanding: that the choice be real. A person who has never been told there is a second path has not chosen the first.

The quiet endings

The other side of this question is dialysis withdrawal, the decision to stop treatment after it has begun. It is more common than most people suppose. Withdrawal is reported as a cause of death in 20 to 30 percent of people with kidney failure in Western countries, and the authors of the reviews I read note that this is probably an underestimate because of inconsistent coding (Source Notes S14). At one Midwestern center that followed 1,226 patients from 2001 to 2015, 536 died, and 262 of those deaths, 49 percent, followed a decision to withdraw. The death-notification forms captured the withdrawal in only about 73 percent of sampled cases (Source Notes S15).

After withdrawal, the end usually comes quickly. In a three-year Italian program that joined nephrology and palliative care, most patients died within a week, and the program reported fewer hospital admissions and more deaths at home (Source Notes S16). An Australian abstract reported that people who stop dialysis typically survive about a week, and that most still die in acute-care hospitals, often against their wishes (Source Notes S17). A week is short. It is enough time to say what needs saying, if the setting allows, and not enough to repair a setting that does not.

I mention these figures not to be grim but because they describe a moment for which most families are wholly unprepared. They arrive at it in a hospital corridor, without a plan, because no one had the conversation months earlier when there was still time to have it well.

What a good conversation sounds like

I have been describing a gap in the abstract, and the gap is easier to feel in a sentence. The conversation that rarely happens might begin something like this: “Before we talk about dialysis, I’d like to understand what matters most to you in the time you have. Some people with kidneys like yours choose dialysis, and some choose to focus on comfort and on staying home. Both are real choices, and I’ll support whichever is yours. Can you tell me what a good week looks like?”

Nothing in that paragraph requires new technology or a new payment code. It requires about ten minutes, a physician who is willing to say that medicine has limits, and a system that does not punish her for taking the time. It also requires the humility to hear the answer. A patient who says that a good week is one in which she can

still garden, and who is then told about a schedule that will take three afternoons, has been given information she can use. A patient who is told only that dialysis is next has not been consulted at all.

The oldest question

Montaigne titled one of his essays with a sentence that has been quoted for four centuries: to philosophize is to learn to die. He did not mean that we should brood. He meant that a person who has looked squarely at the end is freer to choose what to do with the middle. That is the existential point beneath the clinical one. Dialysis is a method of buying time. The seamstress's question, what will it cost me in afternoons, is the right question to ask of any purchase. Some people, asking it, will buy the time gladly and use it well. Some will decide the price is too high. Both are exercising the same dignity.

A conservative reading adds a note of humility. Medicine's power tempts it toward the view that every decline is a problem to be solved. The older and wiser tradition holds that the person is the authority over her own life, that families are the proper witnesses of such decisions, and that the physician's role is to illuminate the choice and not to make it. A system that cannot say "there is another way, and here is how it goes" has not respected the person in front of it.

If you are weighing this decision for yourself or someone you love, please do so with your own clinical team, and with a palliative care specialist if one is available. If you are in distress, call or text 988.

PART TWO
THE DOORS

CHAPTER 6

The Disease No One Feels

“For in much wisdom is much grief: and he that increaseth knowledge increaseth sorrow.”

Ecclesiastes 1:18 (King James Version, 1611)

Kidney disease has an unusual talent for silence. A failing heart announces itself with breathlessness, a failing lung with every flight of stairs. A failing kidney can go about its decline for years without producing a single symptom a person would think to mention. By the time it begins to hurt, in the sense of fatigue and swelling and a taste like metal in the mouth, a great deal has already been lost. Nothing hurts until nearly everything does.

I think of this as the cruelest feature of the disease, and also the most hopeful, because silence is a problem that information can address.

How many, and how few know

The Centers for Disease Control and Prevention estimated, in a national page updated in March 2026, that 14 percent of American adults, about 37 million people, have chronic kidney disease (Centers for Disease Control and Prevention, 2026). The numbers that follow are older, and I give them as context only. An earlier CDC analysis, reported at the time in the trade press, found that 96 percent of people with early-stage disease did not know they had it, and that 48 percent of those with severely reduced function were unaware and not yet on dialysis (Kidney disease affects 1 in 7 Americans, says CDC, n.d.). The American Kidney Fund cites 35.5 million Americans and says nine in ten are unaware, a figure that is widely repeated, though I could not find the calculation behind it (Source Notes S18). A study from the 2020s of patients with moderately to severely reduced function found that roughly half were unaware, and the authors observed that the gap has persisted for nearly two decades (Source Notes S19).

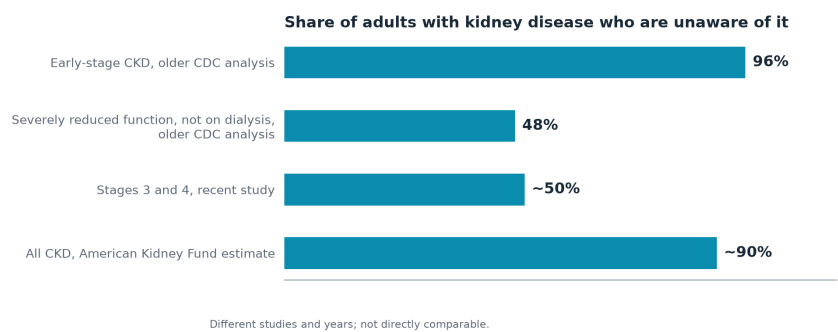


Figure 4.1. Share of adults with chronic kidney disease who are unaware, by group and source. Figures come from different studies and years and are not directly comparable. Sources: Becker’s Hospital Review (n.d.); Source Notes S19.

These figures disagree at the edges, and that is the honest state of knowledge. What they agree about is the shape: a large population, most of it unaware, discovering the disease late. A test that costs little exists. Measuring albumin in the urine is one of the key ways to detect early damage, and yet a study cited in commentary on one major trial found that among adults with type 2 diabetes receiving primary care at 24 U.S. health organizations, the median rate of such testing was only 52.9 percent (Diabetic kidney disease therapies keep flowing, n.d.). The people most likely to be harmed by diabetes were being checked about half the time.

The most hopeful paragraph in the book

If the first half of this chapter is about silence, the second half is about something that has begun to change it. The FLOW trial, a large randomized study, was the first dedicated kidney-outcomes trial of a once-weekly injectable drug of the GLP-1 class, semaglutide, in people with type 2 diabetes and chronic kidney disease. A later analysis found that the benefit held whether or not participants were also taking an SGLT2 inhibitor, a different class of drug that professional guidelines now recommend as first-line treatment for most adults with diabetes and kidney disease (Effects of semaglutide with and without concomitant SGLT2 inhibitor use in participants with type 2 diabetes and chronic kidney disease in the FLOW trial, 2024†).

I will resist the temptation to promise what these drugs will do for the population. I found no study that measured how many eligible patients are actually receiving them, and a promise built on prescriptions nobody has counted is a rumor. But I will say what the pattern implies. For most of the history of kidney medicine, the options on offer began at the point of failure. Now, for the first time,

there are tools aimed at the long quiet stretch before it. The kidney that never fails is the most patient-centered innovation imaginable, since it spares a person the entire apparatus of what comes after.

The chair you never sit in

The second way of avoiding the chair is to skip it. A preemptive transplant takes place before dialysis ever begins. It is the goal many clinicians would choose if they could, and in practice it remains uncommon. One analysis of transplants from 2003 to 2012 found that only 17 percent of recipients overall, and 31 percent of recipients of living-donor kidneys, received one before dialysis. Another reported that only 2.5 percent of people whose kidneys failed in 2015 received a preemptive transplant as their first treatment (Source Notes S20). A 2025 study using national transplant data reminded readers that the survival benefit of very early preemptive transplantation remains debated.

One quiet policy story shows how a single number can decide whether a person gets to skip the chair. For years, a formula used to estimate kidney function included an adjustment based on race, which made the kidneys of Black patients look healthier than they were. Because a transplant candidate's waiting time could begin only when the estimate fell below a threshold, the formula had the effect of starting the clock later for Black candidates. In 2023 the national transplant network required centers to correct this by backdating waiting time for affected candidates, those who would have crossed the threshold on a race-neutral calculation (Organ Procurement and Transplantation Network, n.d.). By July of that year, 6,103 modifications had been completed, with a median of 1.7 years of additional time awarded, and the share of Black candidates

transplanted before dialysis had risen from 11.7 to 13.9 percent, a preliminary finding the authors cautioned might reflect broader trends (6 month waiting time modification and 1 year race-neutral eGFR monitoring report, n.d.). A later quasi-experimental analysis of more than 181,000 adults, reported in March 2026, found that over 21,000 candidates had received added time between January 2023 and June 2025 and that the policy was associated with 5.3 additional transplants per 1,000 listings among Black candidates, in both the pre-dialysis and the on-dialysis groups (Source Notes S21).

I find this story moving for a reason that has nothing to do with politics. It shows that a measuring stick can quietly steal years, that the theft can be undone once someone looks at the stick, and that undoing it gave real people real kidneys sooner. It also illustrates something conservatives understand well: that an institution's rules deserve respect, and that respect includes the willingness to repair them when they turn out to be wrong.

What knowing buys

The book of Ecclesiastes warns that increasing knowledge increases sorrow, and a person told at fifty that her kidneys are failing may well agree. But she is also handed a choice. She can act on the diabetes, the blood pressure, the medications, the diet, the family conversation about donation, the question of whether she would ever want the chair. The silence of kidney disease is not a kindness. It is a theft of time during which choices are still wide. The most patient-centered thing a health system can do is give people the news while they can still do something about it, and give it gently.

CHAPTER 7

The Navigator

“Midway upon the journey of our life / I found myself within a forest dark, / For the straightforward pathway had been lost.”

Dante Alighieri, *The Divine Comedy*, *Inferno*, Canto I (Longfellow translation, 1867)

Picture a woman of fifty-six who has just been told that she may be a candidate for a transplant. The sentence arrives in a conversation lasting nine minutes, and she walks out of the clinic holding a folder. In the folder are six forms, a list of tests, three phone numbers, and a page of instructions written in the dialect of people who work in hospitals. She is told she needs a colonoscopy, a dental clearance, a cardiac stress test, a mammogram, and a social-work evaluation. None of these appointments can be made by the people who requested them. All of them must be completed, in some order, before a committee will consider her.

She is not unintelligent. She kept a small business afloat for twenty years. But she is tired, she lives forty minutes from the center, she works part-time, and the folder sits on the kitchen table, and then on the counter, and then in a drawer.

This is the stretch of the journey that the registries cannot see, because it happens before anyone appears on a list. It is a forest in the poet's sense: the straightforward pathway has been lost. And a number of people have tried to send in guides.

The navigator, tested

The most natural idea is a person whose job is to walk beside the patient through the forest. At Emory University, investigators led by a transplant researcher named Rachel Patzer put that idea to a randomized test. They enrolled 401 patients referred for transplant evaluation and assigned half to a trained navigator, who followed each patient from referral to the decision about waitlisting.

The result was not what the idea's admirers hoped. Overall, waitlisting was not significantly different between the navigated patients, 32 percent, and the controls, 26 percent, and the time from referral to waitlisting was actually 126 days longer for the navigated group. The authors suggested a gentler reading: the navigated patients may have been those with the greatest needs, who required more time for testing and follow-up. In one subgroup, a social-work navigator who helped patients complete tests and attend appointments seemed to increase waitlisting among those who took more than 500 days to get through the process (Source Notes S22).

I find that result more instructive than a victory would have been. It says that a human guide, however well trained, is not a magic key, and that the difficulty of the forest may exceed what one person can supply. It also says something about trial design: a navigator may

bring forward the patients who would otherwise have vanished, and in doing so lengthen the average time, which looks like failure and may be the opposite.

Starting at the dialysis unit instead

A different approach asked a different question. Instead of guiding patients after referral, why not change what happens in the dialysis unit, where most of them spend three afternoons a week? The study called RaDIANT randomized 134 dialysis facilities in Georgia, serving more than 9,000 patients, to receive or not receive an education and engagement program aimed at facility leaders, staff, and patients. Its primary outcome was referral for transplant evaluation within a year. The program substantially improved referral and narrowed racial disparities in it, though the investigators noted that longer follow-up would be needed to know whether the gains carried through to the waiting list. The intervention was later extended to North Carolina and South Carolina (Source Notes S22).

Notice the shape of the success. It did not rely on a single heroic person. It changed an environment: the facility's culture of talking about transplant, the staff's knowledge, the patients' expectations. That fits the pattern I found in Chapter 4, where the most promising programs treated the people near a patient as the unit of change.

Someone who has been there

A third idea is peer mentorship: pairing a patient with someone who has walked the same road. The appeal is obvious. A person who has had a transplant can say things no professional can say. The evidence, however, is mixed. A hospital-based peer mentoring program for

transplant candidates found no significant differences in time to listing or quality of life between those who had mentors and those who did not, and the authors noted that willingness to accept a mentor varied by race (Source Notes S23). A pragmatic randomized trial of 200 hemodialysis patients, reported in the Journal of the American Society of Nephrology in April 2025, measured unplanned hospitalizations and emergency visits over three months; the mentored group's adjusted incidence rate ratio was 0.60 against 0.85 for usual care, but the overall result did not reach statistical significance, and benefit appeared in certain subgroups (Source Notes S23). A feasibility study of 57 patients at a dialysis provider and a trial pairing house-call transplant education with peer mentorship are registered, and I found no outcomes for either.

What this teaches

If you have a tidy mind, this chapter will frustrate you. Navigators did not clearly work, mentors did not clearly work, and the best-supported approach was an educational program that changed the culture of dialysis units. A hasty reader might conclude that the human touch is overrated. I draw nearly the opposite lesson.

What the results suggest is that the human touch is not a single thing. A guide who accompanies a patient through paperwork, a mentor who shares an experience, and a clinic that changes how it speaks are all different interventions, with different mechanisms and different populations, and that none of them has yet been shown to be the key does not mean that none will. It means that this field has not yet done the patient work of learning what kind of help helps whom. I

would want these tested again, with larger samples, with outcomes that include what patients themselves value, and with willingness to publish the disappointments, as Emory's team did.

The woman with the folder deserves more than a folder. She deserves a first conversation that is longer than nine minutes, a single list instead of six, someone who can make the appointments she cannot, a clinic that is not forty minutes away, and, if she wishes it, a person who has been through the forest and is willing to say what the trees look like. Whether any of that will work is a question we can answer. Whether it is worth asking is not.

CHAPTER 8

Dialysis at the Kitchen Table

“Home is the place where, when you have to go there, / They have to take you in.”

Robert Frost, “The Death of the Hired Man” (1914)

Picture a man in his late fifties, standing in his own kitchen at six in the morning, doing something he would have thought impossible a year earlier. He is washing his hands for the third time. Beside the counter sits a machine about the size of a carry-on bag, and he has just threaded a pair of needles into the vein access in his forearm, a task he was taught over four weeks by a nurse who never once made him feel stupid. His wife is making coffee behind him. She has learned to watch the readout, and to hear a small change in the sound.

No such man exists, and for a growing minority of people with kidney failure the scene is simply true.

The slow rise of the house

Home dialysis, which includes peritoneal dialysis done with a catheter and a sterile fluid, and home hemodialysis done with a compact machine, gives people something the clinic cannot: control of the schedule. A person can dialyze at night while sleeping, or for shorter sessions more often, or in the afternoon after the grandchildren leave. Recall what patients named as most important in Chapter 1. Ability to travel, time free from dialysis, and not feeling washed out are the things a home schedule can sometimes give back.

The uptake has been slow. A study using national registry data found that the share of newly diagnosed patients starting on home dialysis rose from 6.8 percent in 2010 to 13.3 percent in 2020, a pace the authors judged too slow to reach the federal goal of having 80 percent of new patients on home dialysis or a transplant by 2025 (Source Notes S24). A 2025 market report counted about 433,400 patients treated in centers and about 78,400 treated at home as of March 31, 2025, which by my arithmetic puts the home share near 15 percent of that combined group, though that is not an official national figure (Matthews, 2025). For home hemodialysis specifically, a manufacturer announcement citing 2022 figures put the share at only 2.4 percent of roughly 550,000 people on dialysis (FDA clears Quanta home hemodialysis system, n.d.).

What the incentives did, and did not do

The federal government tried to nudge the numbers. Under a payment model called ESRD Treatment Choices, dialysis facilities in randomly selected regions, 95 of them, covering 2,591 facilities and 141,890 Medicare fee-for-service beneficiaries, about a third of the

Medicare ESRD population, received payment adjustments tied to rates of home dialysis and transplantation (Centers for Medicare & Medicaid Services, 2025). The first-year evaluation reported about 546 additional patient-months of home-dialysis training, a 9 percent increase, and 225 additional transplants, a 10 percent increase, but no increase in the actual use of home dialysis (CMS releases first-year results of ESRD Treatment Choices model, 2023).

The third-year summary is harder to read. It says that through 2023 the model led to an increase in home-dialysis training and in overall transplant rates, driven by deceased-donor transplants. Elsewhere the same summary says the model did not increase home-dialysis rates, transplant waitlisting, or transplant rates, and that it had no effect on Medicare spending or quality (CMS, 2025). Those statements appear to conflict about transplants, and I could not reach the technical report that would resolve the tension. CMS also proposed in 2025 to end the model early, at the close of that year, and I could not confirm the final decision. What I can say is that financial nudges to facilities moved training more than they moved lives. That is a lesson about where the bottleneck lies. Training is a thing a clinic does. Living at home with a machine is a thing a family does.

The machines that might change the conversation

The dream behind all of this is a device that treats kidney failure without tying a person to a schedule at all. Two efforts show how far the dream is from the kitchen. A wearable artificial kidney, a belt-worn device that filtered blood continuously, was tested in a small human trial. It was well tolerated and cleared waste effectively, but the trial was stopped after the seventh participant because of technical problems with the device, and the authors concluded it would need

redesign (Source Notes S25). At the University of California, San Francisco, a team demonstrated in 2023 that kidney cells housed in an implantable bioreactor could survive in a pig, perform several kidney functions, and avoid provoking an immune attack (Artificial kidney breakthrough as bioreactor survives in pig model without immune response, n.d.). A university web page dated 2026 describes human trials as a realistic goal for 2026 or 2027, but I could not verify that claim and I found no evidence that human trials have begun (Source Notes S26).

I mention these not to raise hopes but to calibrate them. Between a promising pig and a person lies a long road, and honest advocacy should say so.

What a family should be able to ask

If home therapy is to be a real choice, a household needs to be able to put certain questions to the clinic, and to receive answers in plain language. How many weeks of training will there be, and who will be on the phone at three in the morning when an alarm sounds? What happens if the person doing the treatment is sick, or travels, or simply cannot do it one day? Can we try it for a month and go back to the center without penalty? What will change in our house, in water, in storage, in the electric bill, and who covers it? Who checks on the care partner and not only the patient?

None of these is exotic. Each is a question that a thoughtful family would ask before committing to a machine in the kitchen, and a good program would have the answers ready before they are asked. The programs I would trust most are those that treat the household as a partner in a plan and not as a site where treatment is delivered.

The house as the hospital

There is a conservative argument, and a human one, for home dialysis, and they meet at the kitchen table. The conservative argument is that the household is the most trustworthy institution human beings have built, and that policy should support its competence and not replace it. The human argument is that dignity is largely the experience of controlling one's own day.

But the argument has a shadow, and I want to name it. Every hour the patient reclaims from the clinic is an hour handed to the person at the counter. The man in the kitchen has a wife who has learned to hear a change in the machine's sound, and she was never asked whether she wished to learn. A home program that does not count its care partners among its patients has simply moved the burden from the building to the family.

The wise reform, then, is not to push everyone home. It is to make the choice real, to provide the training and support that make home therapy safe, and to ask the whole household what it can carry. Frost's line has a double edge. Home is where they have to take you in. The question is whether we have built the kind of home that can do it.

CHAPTER 9

The Distance Between Here and Care

“There are only two or three human stories, and they go on repeating themselves as fiercely as if they had never happened before.”

Willa Cather, *O Pioneers!* (1913)

In April 2026, a news story from the Nebraska panhandle described a small hospital’s dialysis unit that had closed. Seventeen patients had counted on it, among them a woman named Pieper. Some had moved. One was living in a rental in another city during the week, coming home on weekends. Another was driving more than four hours round trip for each treatment (They counted on a rural dialysis unit to keep them alive. Then it closed., 2026).

I want to stay with that story for a moment, because the distance in it is not a metaphor. A four-hour drive, three times a week, to sit for four hours, and then to drive home feeling washed out, is not a commute. It is a second job on top of the half-time one described in Chapter 1.

How a town loses its machine

The National Kidney Foundation wrote in 2023 that roughly 170 dialysis facilities had closed within the previous year, pointing to staffing shortages, supply chain disruptions, and rising costs (Source Notes S27). Closure is not evenly distributed. A research brief from the South Carolina Rural Health Research Center found that rural dialysis facilities were more likely to be designated low-volume and to run negative margins, and that those at risk of closing offered fewer stations and fewer late shifts (ESRD facility characteristics by rurality and risk of closure, 2017†).

What happens to patients afterward has been studied in older data. A registry-based analysis of closures in small-town, rural, and micropolitan areas between 2001 and 2015 found that care transitions after closure were associated with more hospitalizations, more missed treatments, and a marginal increase in mortality, with the effects more likely among Black patients and those on Medicaid (Source Notes S28). The National Kidney Foundation cited a 2023 study finding that the median one-way increase in travel distance was 6.6 miles in small-town and rural areas, with a wide range, though it noted that the data are old (Source Notes S27). I did not find a current peer-reviewed measurement of closure-driven travel time and missed treatments, which is itself a finding worth stating. The only national picture of how many such closures there have been recently comes from advocacy letters and local news.

The thinning of the people who staff the machines

Behind every dialysis unit stands a nephrologist, and that profession is changing. A commentary from the National Rural Health Association, published in September 2026, reports that the fill rate for adult nephrology fellowship positions fell from 94.1 percent in 2010 to 65.8 percent in 2024, while the number of applicants fell by more than 37 percent. It cites a federal projection of a 21 percent shortage of nephrologists by 2037, and notes that rural areas have about 30 physicians and specialists of all kinds per 100,000 residents against 263 in urban areas (The rural nephrology workforce shortage is already here, 2026). A trade-press report indicated that the 2025 match recovered partway, with a fill rate near 73 percent (Source Notes S29). The commentary is advocacy in tone, and the federal projection should be checked against its original, but the direction of the trend agrees across sources.

Adult nephrology fellowship positions that were filled

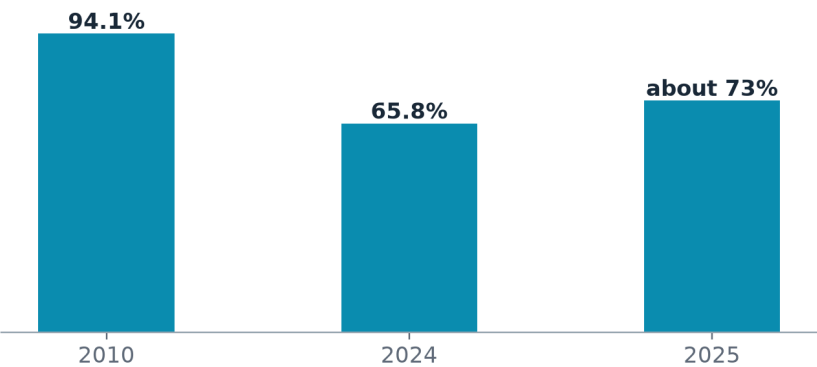


Figure 6.1. Adult nephrology fellowship fill rate, selected years. Sources: National Rural Health Association (2026); Source Notes S29.

An analysis of national data found that the roughly 240,000 rural Americans with kidney failure have less access to nephrology care and higher mortality than urban patients (Source Notes S30). One tele-nephrology program at a rural critical-access hospital reported seeing over 100 patients in its first year, which suggests a possible answer but is a single case report.

The nurse who stays

A dialysis unit is not a machine room. It is a handful of people, a charge nurse, two technicians, a dietitian, a social worker who comes on Thursdays, and what keeps a rural unit open is often nothing more than the willingness of these few to keep working there. The National Kidney Foundation's letter named staffing shortages first among the causes of closure (Source Notes S27). When one nurse leaves a small unit, the unit does not lose one-fifth of its capacity. It may lose the ability to run its late shift, and the patients who work are the ones who needed the late shift. The South Carolina brief's observation that rural units at risk of closure had fewer stations and fewer late shifts shows how the thinning works: it narrows the hours a patient can choose before it takes the unit away altogether.

I mention it because the usual language of workforce policy speaks of numbers, positions, and pipelines. What a rural patient experiences is a face. She knows the technician who learned how her vein behaves. She knows who will call the clinic in the morning if she does not come in. When that person leaves, the loss is a relationship, and relationships are the part of care that dashboards cannot see.

A doctor on a screen

One response to the shortage is telemedicine. A report from a rural critical-access hospital described a tele-nephrology program that saw more than 100 patients in its first twelve months, a single case report and not a trial (Source Notes S30). It is easy to be cynical about this, and also easy to be starry-eyed. A video visit cannot place a needle or place a hand on a swollen ankle. But it can reach a county that has no nephrologist at all, and for the patient who has been choosing between a four-hour drive and no visit, a screen is not a lesser option. It is the only door that is open. The honest approach, as with every intervention in this book, is to test it, and to ask patients whether it gives them back an afternoon.

What I could not find

I looked for a federal count of how many dialysis facilities have closed in recent years, how far patients now travel, and how many missed treatments follow. I did not find one in a form I could cite. The picture comes from advocacy letters, an older registry study, and local news, and I think that gap is itself informative. A country that counts every transplant and every organ cannot say, in a form a reader can find, how many of its dialysis units have shut their doors.

A civic institution

Here I want to make an argument that sounds old-fashioned. A town's dialysis unit is not only a business. It is a civic institution, like the school or the volunteer fire department, whose presence tells the

people who live there that their lives will be held in place. When it closes, the loss exceeds the lost treatments. It is a message about who the country is willing to keep nearby.

Conservative thought has a name for the principle that serves here: subsidiarity, the idea that matters ought to be handled by the smallest, nearest competent authority, and that larger bodies should support the smaller and not displace them. Applied to kidney care, it would suggest that policy should ask what it takes to keep care close, and treat each closure as a failure to be investigated, not a market outcome to be shrugged at. I do not know the right instrument. A payment adjustment for isolated units, a way of sharing nephrologists across counties, a home-dialysis program designed for places where the nearest clinic is two hours away, each is plausible, and none has been tested at the scale needed. What I know is that the question is not being asked with the urgency of the loss.

Cather wrote of the plains that there are only two or three human stories, and they keep repeating. One of them is the story of people who build a place and then watch the services that made it livable move away. The kidney patient driving four hours is a sentence in that story. We could, if we chose, write a different ending.

CHAPTER 10

The Storm

“A prudent man foreseeth the evil, and hideth himself: but the simple pass on, and are punished.”

Proverbs 22:3 (King James Version, 1611)

Consider a house on the Gulf Coast in late August. A storm has been named, tracked, argued about on television, and has swerved. A woman who is on dialysis three times a week looks at her calendar and does what she has trained herself to do. She counts forward. Her next session is Thursday. The storm will arrive Wednesday night. The clinic has called, a recorded voice announcing that it will close if the power fails. She has a packet in a folder, a diet sheet for emergencies, a bag by the door. She also has a daughter two states away who has told her, firmly, to come.

What the storm threatens is not only the roof. A dialysis machine needs three things the way a lung needs air: electricity, clean water in large volumes, and a road that a person can drive on. Take any one away and the hours that Chapter 1 counted stop being a schedule and become a countdown.

What happens when the lights go out

The research on disasters and dialysis is substantial, partly because dialysis patients are among the first people in a disaster whose illness cannot wait. A 2019 systematic review of studies in the Americas found that the loss of electricity, drinkable water, and transportation, and the closure of dialysis facilities, can disrupt care and lead to missed sessions and to more hospitalizations and emergency visits (Source Notes S31).

After Hurricane Katrina in 2005, a survey of patients from nine New Orleans dialysis units found that 16.8 percent missed three or more treatments and 23 percent were hospitalized in the following month. A national registry analysis linked the storm to higher mortality and hospitalization, though the authors cautioned that the makeup of the patient population changed before and after, which can distort comparisons (Source Notes S31).

The most thorough single study came after Hurricane Sandy in 2012. Researchers followed 13,264 in-center dialysis patients in the affected areas and compared them with patients in comparison regions. The rate of emergency visits was 4.1 percent in the storm group against 2.6 and 1.7 percent in the two comparison groups, hospitalization was 4.5 percent against 3.2 and 3.8, and the 30-day death rate was 1.83 percent against 1.47 and 1.60 (Dialysis care and death following Hurricane Sandy, n.d.[†]). It is a modest difference in absolute terms and a meaningful one in a vulnerable population.

The thing that helped

What the Sandy study showed next is the part that every policymaker should memorize. Fifty-nine percent of the patients received early dialysis, an extra treatment before the storm hit, and those who did were less likely to go to an emergency room or be hospitalized within a week, with odds ratios of 0.75 and 0.77. The commentary on the study made the obvious observation that about 40 percent of patients did not receive early dialysis, which means that the preparedness was good and incomplete (What nephrologists learned from Hurricane Sandy, 2015†).

Puerto Rico after Hurricane Maria offers a counterpoint worth taking seriously. A study of mortality there found that deaths among people on dialysis were not higher afterward than they would otherwise have been, and the authors credited adherence to disaster-preparedness guidelines and early government response (Source Notes S31). In a hurricane that devastated the island's infrastructure, a population dependent on electricity and water emerged without excess death. It is not proof that preparation works, but it is an example of what it looks like when it does.

The conservative virtue of foresight

The proverb at the head of this chapter praises the prudent man who sees the evil coming and takes shelter, and condemns the simple, who press on and are punished. It is a counsel of foresight, and it is the least glamorous virtue in public life. Nobody wins an award for the disaster that did not happen.

I think this chapter belongs in a book about time for a reason. Preparedness is the discipline of spending a few hours now so that a person is not stripped of many hours later. A dialysis unit that keeps a generator, that has a protocol for extra treatments before a forecast storm, that tells its patients in advance where to go and whom to call, is spending a modest amount of the system's time to protect a great deal of the patient's. The people who run such units rarely get credit for it.

It also belongs among the questions about distance. The woman on the Gulf Coast may have a daughter two states away. Many people do not. For them the storm is not a matter of choosing the best option but of finding any option, and a system that wishes to take seriously the principle of caring for the near would ask who is within reach of each person on its list when the road is under water.

What the disaster literature leaves open

I did not find recent data on winter storms or the grid failures that now accompany them, though I would expect the same three needs, power, water, and a road, to govern. Nor did I find evidence on how many dialysis facilities in the country have a tested emergency plan. What I found suggests that early dialysis works, that about four in ten patients in one major storm did not get it, and that the cost of preparing is small. That is not a mystery needing more research. It is a gap needing a decision.

CHAPTER 11

The Emergency Chair

“I will use treatment to help the sick according to my ability and judgment, but never with a view to injury and wrong-doing.”

The Hippocratic Oath (Jones translation, 1923)

Picture a man of thirty-eight who does not have a telephone number he can give a doctor. He came to the United States as a young adult and has worked since then in jobs that do not offer insurance. A year ago his kidneys began to fail, and he did not know, because he did not go to a doctor, because doctors cost money he did not have. Now he is in an emergency room, short of breath, his potassium dangerously high. They will dialyze him tonight. They will send him home in a day or two, and he will not be offered a chair again until he becomes sick enough to qualify.

This is a composite, but the arrangement it describes is real, and it is one of the least discussed corners of American kidney care. In some states, people without legal immigration status who have kidney failure can receive dialysis only when they arrive at an emergency department critically ill. It is called emergency-only hemodialysis.

I want to approach it with care, because the subject is politically charged and I do not intend to settle any argument about immigration. My subject is narrower and, I think, one on which people of very different convictions can agree: how well this arrangement works as medicine, and what it costs.

What the comparison shows

The most careful comparison I found followed 211 undocumented patients who began dialysis between 2007 and 2014 at three hospitals. Two of them, in Denver and Houston, offered emergency-only dialysis. The third, in San Francisco, offered standard scheduled dialysis. After adjustment, patients receiving emergency-only treatment had a higher risk of death, with a three-year hazard ratio of 4.96 whose interval was wide enough to include no difference (0.93 to 26.45), and a five-year hazard ratio of 14.13, with an interval wider still (1.24 to 161). They also spent about ten times as many days in acute care as expected (Source Notes S32). The study is observational and the patients differed in ways that adjustment could not fully remove: those who got standard care were more likely to start with a fistula or graft and had better blood counts. The intervals show how little the data can say about the size of the difference. They do not erase its direction.

A second cohort of 181 adults compared scheduled dialysis with emergency-only dialysis. One-year mortality was 3 percent for scheduled treatment and 17 percent for emergency-only. Emergency-only patients made six more emergency-department visits per month, had 1.5 more hospitalizations, and spent ten more days in the hospital over six months, and their care cost \$5,768 more per month (Source Notes S33).

What Colorado changed

In 2019, Colorado decided to count kidney failure as a qualifying condition for emergency Medicaid, which extended coverage to scheduled outpatient dialysis three times a week for those who qualify. A published account reports that before the change, the state's emergency Medicaid dialysis spending for this population was close to \$20,000 a patient per month in 2017 and 2018, and that after the change, the average monthly expenditure per patient was \$5,574 even as the number of patients served rose to about 135 a month (Source Notes S34). That account comes from state data reported in a secondary publication, and the totals it gives are not easy to reconcile with one another, so I rely only on the direction: scheduled care cost less per patient than emergency care.

An argument from prudence

I make no claim about who deserves what. I make a claim about foresight. The conservative tradition prizes prudence, which means attending to consequences, and the evidence I have described points to an arrangement that is deadlier and, per patient, more expensive than the alternative. A system that waits until a person is nearly dead to give him the treatment that would have kept him well has chosen the costliest door.

There is also a plain moral intuition that runs under the practical one, and it is older than any health policy. The oath that physicians have sworn for twenty-five centuries binds them to use treatment for the benefit of the sick. It does not say which sick people. A profession that sends a man home after a rescue knowing he will return in worse condition has not quite honored its oath, though it may have obeyed

its rules. Many nephrologists feel this tension acutely. In the qualitative literature I read, some describe the moral distress of providing emergency-only care as among the hardest parts of their work.

Not the whole story

I should say what I do not know. I found no 2025 study of emergency-only dialysis, and the main cohort studies are from 2018 and rely on data that are now a decade old. I found no outcome data from Texas beyond the early cohort. Policies have shifted in some states since, and some of them may have changed the picture. The evidence is not unanimous about the size of the differences, and the largest estimates carry wide uncertainty.

But the shape is consistent, and it fits what the chapters on time and distance have shown. People who cannot reach the chair on a schedule are not saved money. They are charged later, in worse form, and at a higher price.

I think of the man in the emergency room. He does not know about hazard ratios. He knows that the last time, he was breathless, and that they helped, and that he will be told to come back when it gets bad again. He is waiting in a different kind of line, one whose only way forward is to become sicker. If there is a principle of kidney care in this book, it is that no one should have to earn a chair by nearly dying.

PART THREE
THE GIFT

CHAPTER 12

The Long Chain

“The only gift is a portion of thyself.”

Ralph Waldo Emerson, “Gifts,” *Essays: Second Series* (1844)

Start with two couples who have never met. In the first, a husband needs a kidney and his wife wants to give him one, but their blood types do not match. In the second, the same story, the same love, the same incompatibility, in a different state. Each pair is stuck. Each has a willing donor who cannot help the person she most wants to help.

Then someone does something clever. The first wife’s kidney goes to the second husband. The second wife’s kidney goes to the first husband. Two transplants happen where none could. Nobody has been asked to give to a stranger, and everyone has, without quite noticing.

That is the simplest form of kidney paired donation, and it has become one of the quiet marvels of modern medicine. It is also a puzzle with an almost unbearably human structure, because the longer you let it run, the more it begins to resemble something older than medicine.

The chain that starts with a stranger

Now add a person who walks in with no one in particular to help. A non-directed donor, sometimes called an altruistic donor, gives a kidney to a stranger. That first gift can unlock a pair that had been stuck, whose donor then gives to the next pair, whose donor gives to the next, and on it goes, each link a donor who gives only because someone else gave first. The chain can end by sending its last kidney to a person on the deceased-donor waiting list, so that nobody is left with a debt unpaid.

In 2009, the New England Journal of Medicine described a ten-transplant chain begun by a single altruistic donor in July 2007, coordinated over eight months by two registries, across six hospitals in five states. The authors reported something touching in its plainness: no one reneged (Rees et al., 2009). Every donor, having seen her loved one saved, went through with her own operation, though she had nothing left to gain. Think about that. The chain depends on each link doing a hard thing after the incentive is gone.

Chains have grown longer since. The National Kidney Registry has described one that began with a non-directed donor and ran across 17 hospitals in 11 states over four months, ending with a thirtieth transplant, a record at the time (Source Notes S35). A chain at the University of Alabama at Birmingham that has been reported as the

longest at a single center has been described in different announcements as 74 donors and 74 recipients and as reaching 100 or more recipients (Nation's longest single-site kidney chain reaches 100, n.d.; Source Notes S36). The sources disagree on the current total because they count different things, completed versus ongoing chains, single-center versus multicenter, transplants versus surgeries. I will not pretend to a precision the record lacks.

Scale

What began as a curiosity is now a meaningful share of the field. The National Kidney Registry announced its 10,000th living-donor transplant, with 1,744 facilitated in 2024 alone, and projected that it would account for roughly 30 percent of all living-donor kidney transplants in the United States in 2025. In one recent month, its 104 member centers together facilitated a record 179 transplants, surpassing the 157 set in July 2024 (National Kidney Registry facilitates 10,000th living donor kidney transplant, n.d.). These are the organization's own figures, and they should be read as such, but the trend is not in doubt.

One gift unlocks the next, and nobody repays the person who gave to them

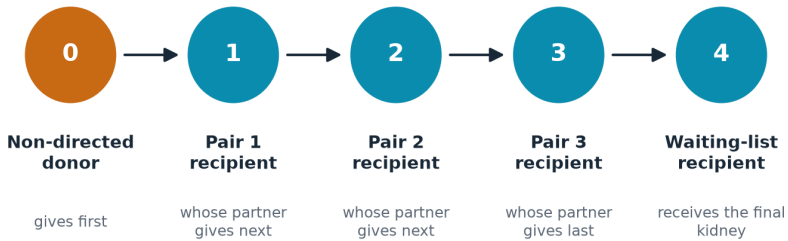


Figure 7.1. How a paired-donation chain works. A non-directed donor starts the chain; each donor gives to the next recipient; the final kidney goes to someone on the waiting list. Schematic, not a real chain.

Why bigger pools help

There is a piece of logic underneath all of this that does not depend on any particular statistic. A pair that cannot match each other has a limited chance of matching with a small group of other pairs. Add more pairs and the chance rises, not by a little but by a lot, because the number of possible combinations grows much faster than the number of participants. This is the reason that a national registry can do what a single hospital cannot, and the reason that the National Kidney Registry's growth matters. Every added pair is a possible key to a lock that someone else has been waiting to open. The advantage compounds, which is why a field built on voluntary gifts can behave, in this one corner, like a network whose value rises with each person who joins.

It is also the reason that every refusal to travel has a cost. A donor who cannot or will not go to another center, a hospital that does not participate, a patient who is never told that this option exists, each subtracts from the pool, and the loss is borne by strangers they will

never meet. The conservative instinct to keep things local is right in a great many parts of this book. Here it meets its limit, and the chain's strength comes from asking local institutions to participate in something larger than themselves.

The promise nobody enforces

One detail of the 2009 chain stays with me more than the transplant counts. Nobody reneged. There is no contract that compels a donor to go through with surgery after her own relative has received a kidney, and in a chain of a few dozen links, the odds that some link will break are not small. The registry's rules can encourage and its coordinators can encourage, but ultimately the chain holds because people keep their word when no one can make them. If you wanted a single example of what a free society runs on, you could do worse. We tend to describe such societies in terms of rights and markets. They also run on promises kept in the dark.

What the chain teaches

I find three lessons in this machinery, and none of them is technical.

The first is that generosity can be organized without being cheapened. The registry's software matches blood types and antibodies, but the heart of the thing is a promise kept by strangers. Emerson wrote that "the only gift is a portion of thyself," and he meant that a gift worthy of the name costs the giver something of her own. A living kidney donor gives a literal portion of herself. A system that honors that must resist treating her as a supply.

The second is that reciprocity, at its best, travels forward and not back. The old moral economy of favor owed and favor repaid was a dyad. The chain makes it a line. You cannot repay the person who gave to your husband, because she gave because someone gave before her. You can only pass it on, and the person you pass it to cannot repay you either. A line of debts that nobody collects is the clearest definition of grace I know.

The third is a conservative one. The chain is built from voluntary acts, and its strength comes from that. Remove the voluntariness, by compulsion or by bribery, and something essential would be lost, though reasonable people differ about where the line falls, as the next chapter will show. What the chain needs from policy is not control but care: reliable logistics, clear information, protection against financial harm to the donors, and a refusal to let the whole thing be turned into a marketplace.

CHAPTER 13

The Receipt

“The cost of a thing is the amount of what I will call life which is required to be exchanged for it, immediately or in the long run.”

Henry David Thoreau, *Walden* (1854)

Imagine a woman who has decided to give a kidney to her brother-in-law. She has been evaluated, approved, and scheduled. Then she sits at her kitchen table with a pencil, the way people do before a big decision, and begins to write down what it will cost.

The medical bills are covered. That part she has been told, and it is true. But there are things the bills do not include. Gas and a motel for the evaluation trips, and for a second trip, and a third. The two weeks she will miss from work, unpaid, since her employer offers no leave she can use. The child care for the week she is in the hospital. The elder care for her mother. The friend who will take the dog. When she is done the list runs to the bottom of the page, and she looks at it for a long while. She will probably give anyway. Not everyone does.

The donor is invented. Her situation belongs to a large number of people who might otherwise give.

A decline with a pattern

The count of living kidney donors in the United States has recovered somewhat from its pandemic dip, but it has not recovered to where it stood before. One news report, drawing on registry data, put the number of living kidney donors at 6,522 in 2025, up from 6,419 in 2024 and 6,290 in 2023, and still below the 6,867 of 2019 (PhillyVoice, 2026†). I classify that as a single-source figure and urge anyone who quotes it to check the registry. An older analysis described a longer decline after a peak in 2004, most pronounced among men, Black adults, and younger and lower-income adults (Source Notes S37). The pattern is not only American. A 2026 article reports that Australia performed 253 living-donor kidney transplants in 2024, down from 354 in 2008, with people from lower socioeconomic groups disadvantaged (Source Notes S38).

When a decline appears in more than one country and concentrates among those with the least cushion, it is reasonable to suspect that the cause is not a failure of kindness. A 2015 consensus conference of transplant professionals reached that conclusion a decade ago, naming the financial burden on donors as a possible source of the decline, and recommending standardized reimbursement for lost wages and incidental costs along with legal protection for employment and insurability (Tushla et al., 2015†).

What changed in 2026

In February 2026 Congress acted. The Honor Our Living Donors Act, part of that year's appropriations law, took away a rule that had always seemed backward: the federal reimbursement program could previously consider the recipient's household income when deciding

whether a donor qualified for help. Under the new law, the recipient's income no longer counts, and the donor's own circumstances determine priority. Eligible donors may receive up to \$6,000 per organ for travel, meals, lost wages, and child and elder care, and the program is now required to report to Congress each year on how much it spent, how many donors it reimbursed, and how many applied without being reimbursed (Source Notes S39).

A physicians' group, in a funding letter, said that 21 percent of donors in the program reach the \$6,000 cap (Source Notes S39). That detail deserves attention. If one in five donors exhausts the ceiling, the ceiling is binding for a meaningful minority, and the woman at the kitchen table with her list may be one of them. In July 2026 the federal health-resources agency requested public comment on the rules (Health Resources and Services Administration, 2026a†). The final shape of the program is still being written.

State law is a patchwork. The American Kidney Fund's report card in May 2026 found gains in several states and little or no progress in many others, and reported that 28 states have laws guarding living donors against insurance discrimination, though protection can be lost if a donor moves (American Kidney Fund, 2026). Private programs fill some gaps, such as wage reimbursement and life insurance offered through the paired-donation registry of the last chapter (Source Notes S40).

The harder question

There is a bolder proposal before Congress, and it deserves a fair hearing. The End Kidney Deaths Act, introduced in 2025, would create a refundable federal tax credit of \$10,000 a year for five years, a total of

\$50,000, for people who donate a kidney to a stranger, and it would sunset after ten years (End Kidney Deaths Act, 2025). It was still in committee as of spring 2026, and a policy tracker rated its chances as low (Source Notes S41).

Honest people are divided, and they are divided for good reasons. The case against is that payment could draw poor people into risks they would otherwise decline, that it might crowd out the altruism that gives a gift its meaning, and that a system built on the idea that organs are given, not sold, might lose public trust if that idea blurred. A federal law, the National Organ Transplant Act of 1984, prohibits the transfer of organs for valuable consideration, and the bill's designers chose a tax credit to respect that principle while changing the incentive (National Organ Transplant Act, 1984). The case for is that the status quo already asks donors to pay, and that the waiting list is a place where people die.

I have a modest view, and it is a conservative one in the sense of caution. Whatever one concludes about payment, no one should object to ending the receipt. A society that asks a person to give an organ to a family member or a stranger should not also ask her to subsidize the operation. Reimbursement that is complete, simple, and not means-tested for the recipient is a debt of ordinary decency. After that, if anyone wishes to test an incentive, it should be tested as an experiment, with outcomes defined in advance and independent evaluation. Policy without evidence is a wager made with other people's lives. Thoreau's accounting is the right one. The price of a thing is the life it costs. The donor deserves to know the price, and to be told it will not be charged to her twice.

CHAPTER 14

The Hallway

“It is better to go to the house of mourning, than to go to the house of feasting: for that is the end of all men; and the living will lay it to his heart.”

Ecclesiastes 7:2 (King James Version, 1611)

Picture a hallway outside an intensive-care unit, late at night. The lights have been dimmed to the amber of a hotel corridor. A family stands at the end of it, close together, not speaking much. Behind a door a person they love is being kept alive by machines, and a doctor has just explained, as gently as such things can be explained, that the person will not recover. A second person has now appeared, quieter and softer-spoken, wearing a lanyard. She is the one who will ask the question.

I have spent much of this book on people whose lives depend on a gift. Here I want to spend time with the people who give it, who are not the recipients and not the donors in any operating-room sense, but the families of those who have died. They are, in the end, the ones on whom the entire deceased-donor system rests, and we know remarkably little about how they fare.

What the research shows, and how thin it is

An interview study of 49 potential donor families in Australia found that no family who agreed to donation regretted the decision, while some who declined did. The process was often long and stressful, but not enough to reduce satisfaction with the outcome (Source Notes S42). A review of the literature reached a subtler conclusion: dissatisfaction with how hospital staff behaved was more likely than the donation itself to worsen families' grief, and the aggravating factors included lack of privacy, a feeling of being pressured, and fear of not honoring the wishes of the person who had died (Source Notes S42).

Support after the fact is uneven. A 15-country European inventory found that about half of respondents thought aftercare for donor families could be improved, and that anonymous written contact between donor families and recipients is possible in almost all countries, with direct contact permitted in only a few (Source Notes S43). An analysis of 1,613 letters over seven years, presented as a conference abstract, found that recipients wrote more often than donor families and usually wrote first, that recipients' letters were mostly about gratitude, and that donor families' letters were mostly about the person who had died (Source Notes S43). A Southern California procurement organization reported that consent rose after it created a family services and aftercare department in 2002, though it is an abstract and cannot establish that the department caused the rise (Source Notes S43).

What strikes me about this literature is how small it is. For a practice that touches thousands of families a year and whose entire legitimacy depends on their trust, there is surprisingly little research

on how those families fare over time. The reviews say as much.

What happened to trust

The trust on which all of this depends has been tested. In the autumn of 2024, news coverage of a Kentucky case, in which a patient was described by his family as showing signs of life as procurement was prepared, prompted a surge of people removing themselves from donor registries. The national registry reported an average of 170 removals a day in the following week, ten times the rate of the same week a year earlier, a count that did not include emailed requests or state registries. The head of the national association of procurement organizations said that organ donation is based on public trust, and that once eroded, it takes years to regain (Associated Press, 2024).

In March 2025 the federal oversight agency completed a report on its investigation of the Kentucky organization. It reviewed about 350 cases in which a planned recovery had been authorized but not completed. In 73 of them it concluded that officials should have reconsidered sooner, because the patients' level of consciousness was high or improving. In at least 28 cases, 8.0 percent, the records showed no documented time of cardiac death, which the agency described as suggesting potential survival to hospital discharge (Health Resources and Services Administration, Division of Transplantation, 2025). Let me be careful in how I say this, because the wording matters. It is a finding about documentation and judgment, serious in its implications, and not a finding that twenty-eight living people underwent removal. The organization has disputed characterizations of its practices, and I treat the most publicized

individual account as an allegation. A summary issued by the Department of Health and Human Services in July 2025 stated that concerns were identified in 103 of 351 cases (Source Notes S44).

The response was substantial. The agency required a root-cause analysis and a policy allowing any staff member to halt a donation, directed the national network to report every pause for a safety concern, and noted that it had received reports of similar patterns elsewhere (Health Resources and Services Administration, Division of Transplantation, 2025). In September 2025 federal officials decertified a Miami procurement organization in mid-cycle, citing unsafe practices, inadequate training, chronic underperformance, understaffing, and paperwork errors, the first action of its kind (Fierce Healthcare, 2025). In 2026 the national network's board adopted new requirements for donation after circulatory death, which the oversight agency described as strengthening protections for potential donors (Health Resources and Services Administration, 2026b).

The fence

G. K. Chesterton left a parable that conservatives are fond of citing, and this seems the right place to cite it. A man comes upon a fence across a road, and a reformer says he does not see the use of it and proposes to clear it away. The wiser reformer answers that if you do not see the use of it, you may not clear it away. Go away and think. When you can come back and tell me you do see the use, I may allow you to destroy it (Chesterton, 1929).

The voluntary donation of organs after death is such a fence. It rests on a delicate, unwritten understanding that those who give are not being used. Every reform, whether a new metric, a new tier, or a

new payment model, is a hand on that fence. The events of 2024 and 2025 are best read as a reminder of why the fence was built. A donation system survives on its reputation for being, first, a system that does no harm.

I do not think the answer is to make donation harder to carry out. I think the answer is to make the process legible to the family in the hallway: who determines death, who is empowered to stop, what happens next, and who will call in a month. The people in the hallway are keeping a promise on behalf of someone who can no longer keep it himself. They deserve a system that behaves as though it knows that.

PART FOUR
THE MACHINERY OF HOPE

CHAPTER 15

Three in the Morning

“The art of being wise is the art of knowing what to overlook.”

William James, *The Principles of Psychology* (1890)

Picture a surgeon asleep in her own bed when the telephone lights up the nightstand. It is 3:07 a.m. A coordinator from an organ procurement organization is on the line, speaking fast and clearly in the manner of people who make this call often. A kidney has become available. The donor is sixty-one. There was a biopsy, and the result is not wonderful and not terrible. The kidney is a few hours from her hospital by air. A patient on her list, a man she has known for three years, is sixth in line. Four centers ahead of her have already said no.

She has perhaps a few minutes to decide, and the decision has several layers. Is the kidney good enough for this man? Can it be here in time? Will his recovery be harder if it arrives late? If it works poorly,

her program's outcome statistics will carry the mark. If she declines, the offer passes to the next center, and perhaps the kidney is used, or perhaps it is not.

The surgeon is invented. The dilemma is exactly the one the data describe.

What goes unused

In 2024, 29.3 percent of the kidneys recovered for transplant in the United States were not transplanted, up from 18.2 percent in 2013 (Scientific Registry of Transplant Recipients & Organ Procurement and Transplantation Network, 2026a). The registry's analysts say "nonuse," not "discard," and the choice is telling, because it concedes that the fate of a recovered kidney is often a decision and not a destiny.

Two details in that report say something about how decisions behave. Kidneys from donors who had a procurement biopsy went unused 40.8 percent of the time, against 6.4 percent for kidneys not biopsied. Selection explains much of that, since surgeons biopsy the kidneys they already doubt. But a biopsy samples a small piece of an organ and is read by a pathologist who may have been woken for the purpose, and a rate that high invites the question of how consistently such readings turn into verdicts (Scientific Registry of Transplant Recipients & Organ Procurement and Transplantation Network, 2026a). The other detail concerns hepatitis C. Nonuse of kidneys from donors who had been exposed but had no active virus fell from 49.5 percent in 2016 to 32.2 percent in 2024, which the report attributes to the spread of drugs that cure the infection. A drug invented elsewhere changed what surgeons were willing to do, and about a third fewer of

those kidneys were wasted. Much of nonuse, in other words, is not a property of the organ. It is a property of what people believe they can safely do.

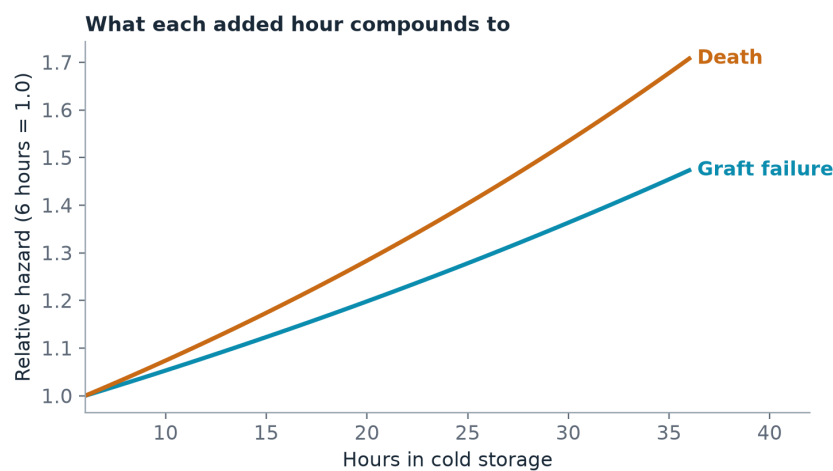
One kidney, many answers

The decision at three in the morning is a human one, and humans vary. A national study of offers made between 2021 and 2023 found that the 193 adult centers accepted a median of 12.5 percent of offers for ordinary kidneys and 7.2 percent for suboptimal ones, with the widest variation among kidneys that had already endured more than 36 hours of cold storage (Quantifying center-level aggressiveness in transplanting suboptimal, deceased donor kidneys in the United States, n.d.[†]). Earlier studies, which I could read only in summary, suggest that variation within a single center can exceed variation between centers, that kidneys are offered a median of seven times before acceptance, and that agreement among transplant nephrologists on whether to accept a given offer was poor, with a statistic of 0.13 where 1.0 means perfect agreement (Source Notes S45).

I want to be fair to the surgeon in the story. Variation among skilled people deciding under uncertainty is not a sign of incompetence. It is a sign that the problem is hard and that the information is thin. The old teaching, put memorably by William James, is that wisdom is largely the art of knowing what to overlook. But to know what to overlook, one has to see everything first, and at three in the morning, nobody does.

The hour as a hidden variable

Every hour on ice has a cost, and the cost compounds. A French cohort of 3,839 deceased-donor recipients found that each added hour of cold storage raised the hazard of graft failure by a factor of 1.013 and of death by 1.018 (Debout et al., 2015†). Compounded across the twenty-four hours between six and thirty, my own arithmetic gives a 36 percent higher hazard of graft failure and 53 percent higher hazard of death. The authors, rounding differently, wrote that a thirty-hour kidney carried a risk near forty percent higher than a six-hour kidney. A study reported in June 2026 of 74 kidney pairs from older donors found that cold storage beyond twelve hours raised the odds of delayed graft function more than sixfold, though it was small and from a single center (Source Notes S46).



Extrapolated from published per-hour hazard ratios (1.013 and 1.018); author calculation, illustrative.

Figure 10.1. Relative hazard of graft failure and death by hours of cold storage, extrapolated from published per-hour ratios (Debout et al., 2015†). Author’s calculation; illustrative only.

The point is not that long journeys doom a kidney. The harm is a slope, not a cliff, and the surgeon in the story is weighing a slope. The deeper point is that the hours are a product of logistics as much as medicine. In 2025 a federal working group delivered a final report with 20 recommendations on moving organs, and the Federal Aviation Administration launched a website to help those who carry them by air, with updated guidance on keeping organs in the cabin (Health Resources and Services Administration, n.d.). A 2024 industry briefing noted that the contract then in force could not require organs to be tracked, which made it hard to learn what happened to those that arrived late or damaged, and the organization long contracted to run the network says it built its own tracking tool, now used by 16 procurement organizations (Source Notes S47). When the national kidney-allocation system was changed in 2021 to share organs more broadly, a federal monitoring report found that kidneys traveled farther, mostly within about 250 nautical miles, and that nonuse rose from 21 to 26 percent (Source Notes S47). A proposal for a more flexible system of ranking candidates, called continuous distribution, was listed as paused as of July 2025, and I found no newer status (Source Notes S47).

What would help her

I do not think the answer is to take the decision away from the surgeon. The answer is to give her a better map than the one she has. A short, standard summary of the donor and the kidney that all centers see in the same format. A projection of what the expected cold time will be, updated as the plane takes off. A way to ask for a second

look at a kidney that has been declined repeatedly. A published, plain-language record of how often decisions like hers turn out well, so that she is not carrying the full weight of uncertainty alone.

Above all I would give her a friend, a colleague or a system that tells her at three in the morning what the evidence says about kidneys like this one, as clearly as a pilot is told what the instruments say. Judgment is a human quality, and it improves with good instruments. That is not an argument against the surgeon. It is an argument for her.

CHAPTER 16

The Odd Machines

“Why should we be in such desperate haste to succeed, and in such desperate enterprises?”

Henry David Thoreau, *Walden* (1854)

The oddest part of transplant medicine is how little of it looks like the future. Much of the daily work still depends on a cooler, a courier, and a telephone. And yet around the edges of that old routine, a set of strange new machines is being built, some of them so unlikely that when you first hear of them they sound like fiction. A pig kidney that works in a human being. A chemical that carries oxygen without a single red cell. A pump that keeps an organ alive outside the body.

Any honest tour of these has to begin by admitting that most promising technologies in medicine fail, and that hope is cheap. So I will do what I have tried to do throughout, which is to ask of each machine a single question: what would it give back, and how well do we know?

The pump

Machine perfusion, the practice of pumping cold preservation fluid through a kidney in place of letting it sit on ice, is the senior member of the family. A randomized trial of 672 recipients, in which one kidney from each donor went to each approach, found that perfusion reduced delayed graft function and improved one-year graft survival (Moers et al., 2009). In November 2025 the manufacturer of one such device announced a ten-year follow-up reporting a 27 percent lower risk of graft failure, with ten-year data for 92 percent of recipients (Organ Recovery Systems, 2025). I classify that as preliminary until it is confirmed by the paper itself, since the announcer sells the device.

Warming the kidney to body temperature while perfusing it has been less persuasive. A randomized trial of kidneys from donors after circulatory death found the approach safe and feasible but no better at preventing delayed graft function, with rates of 60.7 and 58.5 percent (Hosgood et al., 2023[†]). A 2026 single-center trial that added two hours of warm perfusion to cold perfusion found immediate graft function in 39 percent of treated kidneys against 49 percent of controls, with an interval so wide, 0.28 to 1.64, that it could not rule out benefit or harm (Normothermic versus hypothermic machine perfusion in kidney transplantation: A randomized controlled trial, 2026[†]). A 2025 review concluded that adding oxygen during cold perfusion has produced mixed results, and that the benefit of warm perfusion remains unproven (Nature Reviews Nephrology, 2025[†]). Untested is not the same as useless. It is untested.

The molecule

The idea behind oxygen carriers is to add to the perfusion fluid a molecule that does the job red cells do, carrying oxygen, so that an organ cut off from circulation is not starved. A French group tested one such product, extracted from a marine worm and known as HEMO2life, in 60 transplanted kidneys at six centers, comparing them with the paired kidneys from the same donors. The investigators reported higher patient survival at 48 months, 98.3 percent against 86 percent, and the manufacturer reports lower delayed graft function in some subgroups (Source Notes S48). I would treat these claims with unusual caution. It is surprising for a preservation additive to change how long a recipient lives four years later, and surprising results are usually the start of an investigation. The design was not randomized, and the most favorable figures come from interested parties. A randomized trial is registered, and I found no results (ClinicalTrials.gov, n.d.).

The Cold Ischemia Foundation has published educational material about one company in this space, BHOC Therapeutics, and I owe the reader the Foundation's own stated position, which is stricter than any critic's. The company describes a cell-free hemoglobin carrier designed to work across blood groups. Its transplant program says plainly that it is a research concept, not an approved treatment, and that it has not been shown to prevent rejection or to change transplant outcomes (Cold Ischemia Foundation, n.d.-a). The Foundation's evidence brief notes that no oxygen carrier has full regulatory approval for intravenous use in humans (Cold Ischemia Foundation, n.d.-b). Interest is warranted. Promotion is not.

The pig

The most startling development of the past two years is that a gene-edited pig kidney has worked, for months, inside a living person. In January 2025, a 66-year-old man at Massachusetts General Hospital, Tim Andrews, received a kidney from a pig engineered by the company eGenesis. He lived without dialysis for 271 days. An early episode of rejection, driven by T cells, responded to treatment, and no pig virus was detected. The kidney was removed in October 2025 after progressive dysfunction. In January 2026 he received a donated human kidney, becoming the first patient to move from a pig kidney to a human one. The case was reported in *The Lancet* on 3 September 2026 (Patient who received pig kidney becomes first to progress to human kidney transplant, 2026; Immune profiling in a living human recipient of a gene-edited pig kidney, 2025).

The company reported in September 2026 that five patients had received its kidney under an expanded-access program, that three had maintained kidney function for more than eight months without dialysis, that one had passed nine months, a record, and that two had gone on to human kidneys (eGenesis, 2026[†]). That update is a company statement and not peer-reviewed. The first formal trials are underway: United Therapeutics' gene-edited kidney, with ten genetic changes, was cleared by the Food and Drug Administration in February 2025 and its first trial transplant took place at NYU Langone on 3 November 2025 (United Therapeutics Corporation, 2025). The eGenesis kidney was cleared for a formal trial in September 2025, designed for dialysis-dependent adults aged 50 and older who are on the waiting list. I found no patient outcome data from either formal

trial. A recent review in the American Journal of Transplantation takes stock of where the field stands (Xenotransplantation in living human recipients: Where we stand in 2026, 2026).

What each would give back

Machine	What it might return	How well do we know
Cold perfusion pump	Better early function and, perhaps, longer graft life	One large trial; ten-year follow-up reported by the manufacturer
Warm perfusion	A longer, safer time window for the organ	Two randomized trials; no benefit shown on main endpoints
Oxygen carriers	Organs better protected during the journey	Early human data, mostly from interested parties; research stage
Pig kidney	An organ on demand for people unlikely to get a human one	First formal trials under way; a handful of patients
Implantable or wearable artificial kidney	Freedom from the schedule	Animal studies; one small human device trial stopped early

The patience a machine requires

Thoreau asked why we should be in such desperate haste to succeed, and in such desperate enterprises. I read the question not as a counsel of delay but as a caution about motive. The people waiting have every reason to be desperate, and the companies and clinicians who build these machines have every reason to promise. A desperate patient and

an eager inventor, left alone together, can produce either a miracle or a disaster. What separates the two is the dull discipline of evidence: trials designed in advance, results reported whatever they show, independent replication of what manufacturers announce.

Each of these machines, if it works, would return something the person in the chair counts as precious. Time. A quicker graft, a longer one, a pig kidney that needs no wait, a device that frees the schedule. I would not slow any of them down. I would ask only that we measure what they give back, in afternoons, and say so honestly when the answer is not yet.

CHAPTER 17

A Kidney Is Not Forever

“Hope is itself a species of happiness, and, perhaps, the chief happiness which this world affords.”

Samuel Johnson, *The Rambler*, No. 2 (1750)

There is a sentence people say to one another at transplant celebrations, and it is mostly true and slightly dangerous. “You’ve got your life back.” It is true that a working kidney returns something enormous. It is dangerous because it leaves out the next sentence, which is that the kidney has a lifespan, and so, in a sense, does the return.

Consider a thirty-four-year-old woman, two years past her transplant, who has never felt better. She runs on Saturdays now. She is planning a trip. And on some quiet nights, at the edge of sleep, she does a calculation she has told no one about. Her kidney came from a deceased donor. She knows, in the way you know a fact you have chosen not to look at, that it will not last her whole life.

The arithmetic of a graft

An analysis of national registry data, published in the American Journal of Transplantation in 2021, estimated that median graft survival for deceased-donor kidneys rose from 8.2 years for transplants performed in 1995 to 1999 to about 11.7 years in the most recent era, and that for living-donor kidneys it rose from 12.1 years to about 19.2 years for transplants in 2014 to 2017 (Source Notes S49). Those are medians, which means half of grafts last less. A conference abstract using the same registry reported projections for living-donor kidneys of well over twenty years, and noted that five-year and ten-year survival had improved too (Source Notes S49). Both sources use projections for the newest cohorts, and they disagree somewhat with each other, so I treat the figures as approximate.

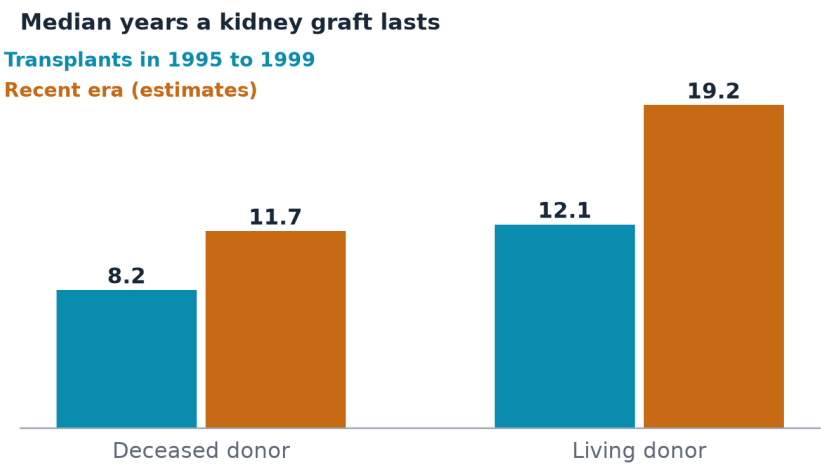


Figure 12.1. Median kidney graft survival by era, deceased and living donors. Source: American Journal of Transplantation registry analysis (2021+); recent-era values are estimates.

The progress is real, and it is a quiet triumph. But look at the implication for a young recipient. If a kidney lasts a dozen years and a person lives sixty more, she will need more than one. The transplant is not an ending. It is the first chapter of a long relationship with a kind of medicine that she will have to renew. A second transplant means a second wait, and the waiting list, as the registry reminds us, is long.

What a transplant buys

It helps to see the larger comparison before looking at the limits. Among patients who began treatment in 2007, national registry data showed an adjusted five-year survival of about 40.4 percent for those on dialysis, 73.7 percent for those with a deceased-donor transplant, and 87.0 percent for those with a living-donor transplant (United States Renal Data System, 2014). Those are old cohorts, and they are not a fair experiment, since people who receive transplants are selected partly for being healthier. A landmark study from the late 1990s, which tried to correct for selection by comparing transplant recipients with others who were also on the waiting list, found that the long-term death rate was 3.8 per 100 patient-years among recipients, against 6.3 among those still waiting and 16.1 among all dialysis patients (Wolfe et al., 1999). The direction is not in doubt. A transplant buys years, and the best estimates say it buys a great many.

That is why the length of a graft matters so much. The years a transplant buys are bounded by the years the graft lasts, and a person who outlives her kidney returns to a waiting list that, in 2024, held 143,982 adults (Scientific Registry of Transplant Recipients & Organ

Procurement and Transplantation Network, 2026b). It is a fact that sits uneasily beside the celebratory sentence, and I think it should be said at the celebration.

The drugs that hold it in place

A transplant works because the recipient takes drugs that quiet the immune system's natural wish to destroy a foreign organ. They are lifesaving, and they are not free of cost. They raise the risk of infection and of certain cancers, they must be taken exactly and for life, and they must be paid for. The previous generation of recipients simply accepted that bargain. The next might not need to.

In a clinical trial of an experimental cell therapy called MDR-101, recipients of kidneys from HLA-matched living siblings received an infusion of cells from the donor along with a conditioning regimen, in an attempt to teach the immune system to accept the kidney as its own. In a phase 3 study with 20 treated patients and 10 controls, interim results in 2023 showed that 16 of 19 patients were off immunosuppression at two years. A report in the American Journal of Transplantation in July 2025 described about 95 percent of the treated patients stopping all immunosuppression around a year after transplant, and 75 percent remaining free of it for more than two years. Four patients resumed drugs, for reasons including recurrent kidney disease and rejection, and none developed graft-versus-host disease (Source Notes S50). The sample is small, restricted to matched siblings, and the figures come from different time points, and I found no 2026 news on regulatory approval.

A transplant physician interviewed in April 2026 described the broader field as moving toward cell therapies, with trials combining stem cells and regulatory T cells in living-donor recipients and a randomized study of whether patients can be safely moved to a single immunosuppressive drug (Transplant tolerance: The frontier of kidney care, 2026†). If it succeeds, the result would be something transplant medicine has chased for sixty years: a kidney that the body treats as its own, which is to say tolerance.

What tolerance means

The word “tolerance” is a clinical term, and I want to take it back for a moment. To tolerate something is to live peacefully alongside it. A person with a tolerated kidney does not carry it as a visitor to be guarded against. She carries it as part of herself. There is a quiet existential difference between taking a drug every day to prevent a rejection and living without that daily reminder that the body is in a negotiation.

Samuel Johnson wrote that hope is itself a species of happiness. He also warned, in the sentence that follows the one most often quoted, that the excesses of hope must be paid for in pain. The woman running on Saturdays knows both halves. A transplant is a gift of years, not a gift of forever. The right response is not to dampen her hope but to meet it with honesty. Tell her the numbers. Tell her about the second chapter. Tell her what is being built for it. And let her go running.

CHAPTER 18

The Handoff

“Rejoice, O young man, in thy youth; and let thy heart cheer thee in the days of thy youth.”

Ecclesiastes 11:9 (King James Version, 1611)

Imagine a sixteen-year-old in a clinic room, sitting on the paper-covered table with her sneakers dangling. She has had her kidney since she was nine. She has been in this clinic so long that the nurses have watched her grow, and the children’s hospital has a mural on the wall that she painted a corner of when she was twelve. She knows her medication names the way she knows her phone number. And she is, this month, being told that at eighteen she will move to the adult clinic, across town, with different doctors, in a building with no mural.

She does not say anything. She nods. She will pay attention to the new place, she tells herself. But she is sixteen, and she has a boyfriend, and a part-time job, and a way of forgetting what she would rather not remember, and the pills are the one thing in her life she has always been told she must not forget.

The girl is invented. The danger she faces is among the best documented in transplant medicine, and among the least discussed outside it.

The age at which kidneys are lost

A large analysis of United States registry data on first kidney transplants from 1987 to 2010, published in JAMA Internal Medicine in 2013, found that recipients aged 14 to 16 had the highest risk of graft loss of any age group, starting a year after transplant and widening at three, five, and ten years. Black adolescents were at disproportionate risk. The authors recommended structured transition programs (Source Notes S51). The data are old now, ending in 2010, but the pattern has been confirmed in other places. A study of national data in England found that young adults lose their transplants and return to dialysis at a far higher rate than people of other ages, and linked this to missed medication and missed appointments (Source Notes S51).

The mechanism is not a mystery, though it is easy to moralize about it. Teenagers forget things, push against authority, want to be like their friends, and sometimes stop taking drugs that make them feel unlike their friends. The drugs carry side effects that matter when you are fifteen: changes in appearance, in weight, in hair. And the transplant itself, being invisible and not painful, is easy to forget until it fails.

A question about systems, not about teenagers

What interests me is where the risk concentrates: not in the teen years as such but at the point when care changes hands. It is the handoff. In pediatric centers a family is surrounded by a team that

knows everyone's name and checks in by phone. In adult centers, a young person is expected to manage her own appointments, her own refills, and her own relationship with a physician she has met twice. In the space between those worlds, a kidney can be lost.

The obvious remedy, a structured program that eases the transition, has been tested, and the result humbles anyone who thought the problem simple. A randomized trial in Germany, called TransNephro, enrolled 102 adolescents and young adults and assigned them to a structured program, which included a central case manager, a communication app, and joint transition rounds for a year before and after transfer, or to standard care. The primary outcome was adherence, measured by the variability of drug levels in the blood. There was no difference between the groups (Source Notes S52). Graft survival was a secondary outcome, and the results I could reach did not report it. A single-center program at Stanford that begins preparing patients for transition at age twelve reported preserving low rates of graft loss while transferring about twice as many patients, though that was a before-and-after comparison and is weaker evidence (Source Notes S52).

The shortage behind the handoff

There is a second, less visible problem at the handoff. Pediatric nephrology is a small and fragile specialty. A report in the trade press said that between 2014 and 2022 the fellowship fill rate in pediatric nephrology was 66 percent, the lowest of all fifteen pediatric subspecialties (The pediatric nephrology workforce is in crisis, 2025). When there are few pediatric kidney specialists, each handoff is also a handoff between scarce hands.

What would a gentler handoff look like?

I do not know. The trial in Germany did not find that a more intensive program improved adherence, and I would not claim that any specific program works. But the right response to a negative result is not to give up. It is to ask what was missing. The program tested was structured and professional. What it may not have supplied is the thing that Chapter 4 found at the heart of the best interventions: the near people. A teenager who has a friend, an older cousin, a young adult who has been through the same thing, someone who is not a clinician and not a parent, might behave differently from one who has a case manager. That is a hypothesis, and not a finding, and it deserves a trial.

Ecclesiastes tells the young man to rejoice in his youth. I do not read it as an instruction to be careless. I read it as a recognition that a young person is entitled to a life that is not defined by her disease, and that a system that asks her to carry her kidney alone, at precisely the age when she most wants to be carried, is asking something that few adults could manage. A better handoff would not ask her to be mature earlier than she is. It would let her be sixteen, and keep her kidney anyway.

CHAPTER 19

The Pharmacy Counter

“The rich ruleth over the poor, and the borrower is servant to the lender.”

Proverbs 22:7 (King James Version, 1611)

Stand behind a man in a pharmacy line on the first of the month, holding a paper bag he has not yet been allowed to take home. The pharmacist is explaining, gently, that the price has changed. His coverage reset. He is not a careless man. He keeps a spreadsheet. He knows exactly how much he has left after rent, and he knows that the pills in the bag are the only thing standing between his transplanted kidney and the machine he left behind.

The man is invented. The gate he faces is one that almost no one thinks of as a gate: the price of the pills that keep a graft alive.

The cliff that was repaired

For decades, a transplant recipient whose only qualifying condition was kidney failure lost Medicare coverage 36 months after surgery, unless disabled or over sixty-five. The anti-rejection drugs did not become cheaper on the thirty-seventh month. Since December 2022 a

benefit called Medicare Part B-ID has covered FDA-approved immunosuppressive drugs indefinitely, for a premium that was \$97.10 a month in 2023 (Source Notes S53). A federal analysis in 2019 had observed that, for patients who lost coverage and had no other source, the cost of continuing therapy could be prohibitive. Congressional testimony cited a report that nearly 70 percent of transplant programs had experienced a death or a lost transplant because a patient could not afford the drugs, a figure that reached me only through that testimony (Source Notes S53). A cost-effectiveness analysis found that extending coverage saved Medicare money while improving outcomes. It is one of those rare policies in which compassion and arithmetic agreed, and it took years to enact.

I could find no peer-reviewed study of how the new benefit has changed patients' out-of-pocket burden or their financial distress after the thirty-sixth month. A repair whose effect nobody has measured has not been finished.

The new uncertainty

The next risk is not a cliff that has been built but one that has been forecast, and I want to describe it with restraint. The 2025 federal budget reconciliation law changed Medicaid in ways that advocacy organizations expect to matter to kidney patients. The National Kidney Foundation, citing the Congressional Budget Office, said that more than 11 million people would lose Medicaid coverage and that more than \$700 billion in funding would be removed over the budget period. It reports that nearly half of dialysis patients rely on Medicaid (National Kidney Foundation, 2025). A nephrology society newsletter noted that a transplant can cost more than \$400,000 and that immunosuppressive drugs afterward cost \$10,000 to \$14,000 a year

(American Society of Nephrology, 2025). A fact brief in August 2026 said that work requirements of 80 hours a month are scheduled to begin in January 2027, with exceptions for some groups, and that eligibility must be rechecked every six months starting by December 2026 (Gigafact, 2026†).

What I do not know is how many dialysis or transplant patients will lose coverage, whether exceptions will protect them, or how immunosuppressant coverage will fare. These are forecasts from advocacy sources and one fact brief, and the enacted text should be read directly. I say so because it would be easy, and wrong, to write a sentence that sounds like a prophecy.

A conversation to have at month thirty

Nothing in the literature I read describes it, but the structure of the problem suggests a practical rule. Every recipient, at about thirty months after surgery, should be asked what coverage she will have at thirty-six, and what the drugs will cost her then. Insurance changes are rarely a surprise to the insurer, and they should not be a surprise to the person whose graft depends on them. A transplant program that made that conversation a routine of the second year, with a social worker or a pharmacist, would be doing the unglamorous work that keeps a gift from being wasted.

The same logic applies at every transition: a move, a job loss, a divorce, a change in a state's rules. Each is a moment at which a person might quietly fall between two forms of coverage, and in a system as complicated as ours the people most likely to fall are those

without anyone to help them read the forms. We spend extraordinary effort delivering the kidney. We spend almost none on delivering the second year.

Independence is a medical outcome

There is a conservative principle in this chapter that has nothing to do with the budget. Proverbs observes that the borrower is servant to the lender. A person who must ask for permission, every month, to take the medicine that keeps her kidney alive is not fully free, and she knows it. The dignity of self-reliance, which conservatives rightly honor, is poorly served by a system that sets a person up to depend on institutions she cannot influence. Whatever one's view of the proper size of public programs, an arrangement that makes the continuity of a transplant depend on a bureaucratic reset date is not a conservative arrangement. It is merely fragile.

The remedies are not exotic. Measure the problem, by asking recipients each year whether they have missed doses for cost. Offer a free navigation call before the thirty-sixth month and before any change in coverage. And design coverage rules with an explicit exception for people whose survival depends on a daily drug, because the cost of that exception is trivial beside the cost of a lost graft.

CHAPTER 20

Who Is Keeping Count

“One of the essential qualities of the clinician is interest in humanity, for the secret of the care of the patient is in caring for the patient.”

Francis W. Peabody, “The Care of the Patient,” JAMA (1927)

Imagine an administrator, late on a Thursday, scrolling through a dashboard with forty-one metrics. Door-to-needle times, readmission rates, graft survival at one year, transplant volumes, compliance with the latest policy. The numbers are carefully defined and the colors mostly green. Nothing on the screen records that on Tuesday a woman named in no database sat for four hours, felt washed out for three more, and missed her granddaughter’s recital.

That is not a failure of the dashboard. It is a description of it. A measurement system is a theory of what matters, and it is worth reading a system’s metrics the way you would read a person’s calendar.

What is counted

Consider the instruments that now govern American kidney care. A payment model for dialysis facilities, the one I described in Chapter 8, rewards home dialysis and transplantation. A newer model for transplant hospitals, which began on 1 July 2025 and runs to mid-2031, scores each participating hospital out of 100: 60 points for the number of transplants it performs, 20 for how readily it accepts organ offers, and 20 for how long its grafts survive. A hospital can earn up to \$15,000 per Medicare kidney transplant and be charged at most \$2,000, and in the first year there is no downside risk at all. As of this writing no first-year results have been published (Centers for Medicare & Medicaid Services, n.d.). A third instrument sorts the nation's 55 organ procurement organizations into tiers on the basis of two rates, a donation rate and a transplant rate, with the lowest tier facing loss of its territory. The first recertification cycle ended in July 2026, a final rule is expected in late 2026, and contracts of organizations in the lowest tier remain in force through 31 January 2027 (Holland & Knight, 2026). I found no published roster of which organizations those are.

The governing body of the whole network is in the middle of rebuilding itself. In April 2026 its board extended its president's and vice president's terms by a year, saying that contractor transitions and funding interruptions had impaired its ability to move initiatives forward (Health Resources and Services Administration, 2026c). It later raised the fee charged for each registered candidate to \$1,268 for fiscal year 2027, about 22 percent more than the year before, and ended one of its standing committees (Health Resources and Services

Administration, 2026d). All of this comes from the agency’s own summaries, and I found no independent press coverage that tested them.

What patients named

Now set that list beside the one in Chapter 1. When patients and caregivers were asked what mattered, they rated the ability to travel, time free from dialysis, the adequacy of treatment, and feeling washed out higher than clinicians did (Developing a set of core outcomes for trials in hemodialysis, 2017†). Read side by side, the two lists barely touch. The system counts volume, acceptance, survival, and compliance. The patients named travel, time, and how they feel in the evening.

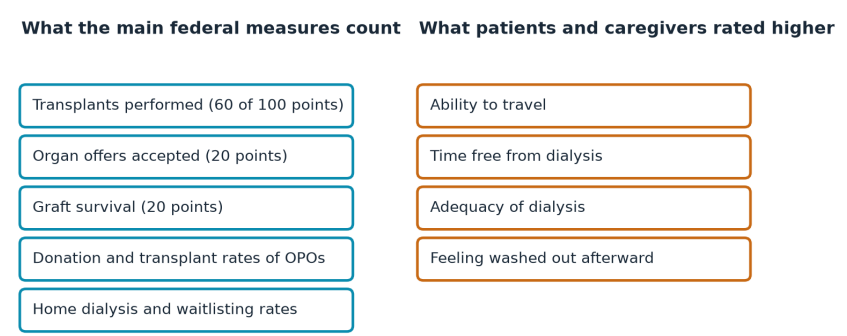


Figure 14.1. What the main federal measures count, and what patients and caregivers rated higher than clinicians. Sources: CMS (n.d.); Holland & Knight (2026); SONG-HD survey (2017†).

I do not say that the system’s measures are wrong. Volume matters, survival matters, and holding monopolies to account matters. I say that the instruments are incomplete in a specific way, and that the incompleteness lands on the people with the most at stake.

The idea behind a core outcome set

The international survey in Chapter 1 belonged to a larger movement within nephrology, one that deserves more credit than it gets. The premise of a core outcome set is simple and slightly radical: before researchers run a trial, they should agree, with patients in the room, on the short list of outcomes that every trial in a field ought to measure. For decades each trial chose its own, usually the ones that were easiest to count, so that results could not be compared and the things patients cared about were often not measured at all. A core outcome set is an attempt to correct that, by writing the patient's priorities into the design of research itself.

I find this encouraging, because it shows that the problem described in this chapter has been recognized by people inside the field, and because it supplies a model for the dashboard I am proposing. If a research community can convene more than a thousand patients, caregivers, and clinicians in 73 countries to agree on what trials should count, a health system could convene a few hundred of its own patients to say what its dashboard should count. The hard part is not the method. It is the willingness to be surprised by the answer.

The law of the good metric

The anthropologist Marilyn Strathern put a warning in one sentence that has since been widely repeated: when a measure becomes a target, it ceases to be a good measure (Strathern, 1997). Every metric in this chapter is vulnerable to it. A volume score can be raised by transplanting those who would have done well anyway. A graft-survival score can be protected by declining difficult cases. An

acceptance score can be improved by accepting what should be declined. The remedy is not to abandon measurement, which would be a disaster for the people who most need it. It is to measure in groups, so that anyone who inflates one number reveals themselves in another, and to include in the group the things that cannot be gamed because they are reported by the patient.

A plain way to tell noise from news

Industry has a small tool for exactly this problem, called a control chart. It plots a measure over time and draws bands around its ordinary range, so that you can tell a real change from the normal wobble. Imagine applying it to the share of recovered kidneys that go unused. That figure was 18.2 percent in 2013, 26.6 percent in 2022, 27.9 percent in 2023, and 29.3 percent in 2024 (Scientific Registry of Transplant Recipients & Organ Procurement and Transplantation Network, 2025, 2026a). No chart is needed to see that this is not wobble. But the nation publishes such numbers once a year, years late, and a number published once a year cannot reveal a drift that begins in March.

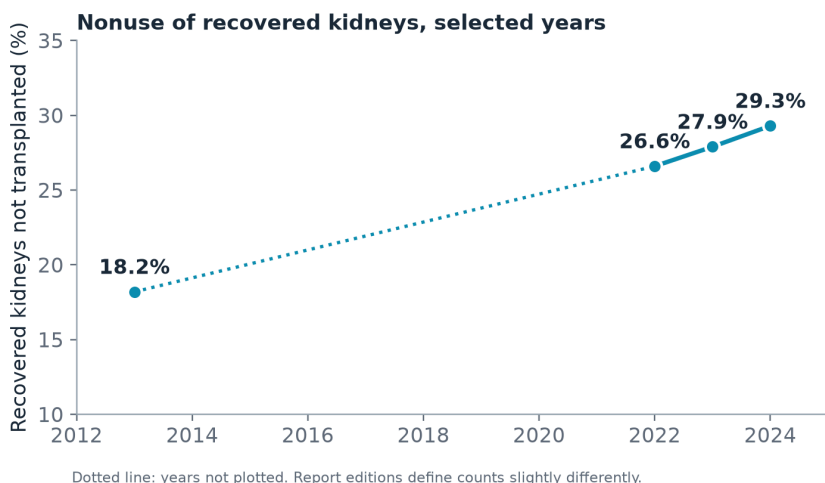


Figure 14.2. Share of recovered kidneys not transplanted, selected years. Definitions differ slightly between report editions. Sources: OPTN/SRTR annual data reports (2025, 2026a).

A dashboard a patient could read

Suppose we began over. I would ask a few hundred patients and care partners to name the five things they most wanted measured, and I would measure those. Time in the chair and in recovery. Distance traveled. Days at home. How much the care partner is carrying. Whether medicine was ever skipped for cost. I would publish them monthly and plainly. Peabody wrote in 1927 that the secret of the care of the patient is in caring for the patient, and the sentence is easy to quote and hard to operationalize. A dashboard that begins with the patient's own list is as close as I know to a way of doing it.

CHAPTER 21

What a Conservative Would Keep

“If you don’t see the use of it, I certainly won’t let you clear it away. Go away and think. Then, when you can come back and tell me that you do see the use of it, I may allow you to destroy it.”

G. K. Chesterton, *The Thing* (1929)

I have called this book’s second register conservative, and I owe the reader an account of what I mean, because the word has drifted a long way from its roots. A conservative, in the tradition I have in mind, is not someone who opposes change. Burke insisted that a state without the means of some change is without the means of its own conservation. A conservative is someone who asks, before changing a thing, what it is for.

Seen that way, the American kidney ecosystem looks different than it does to either its defenders or its critics. It is an assembly of institutions, most of them remarkable, some of them decaying, a few of them invisible until they vanish. The conservative’s first duty is to inventory what is worth keeping.

What I would keep

The gift. The entire edifice rests on the free decision of families to give, and of living people to part with a piece of themselves.

Chesterton, in his book *Orthodoxy*, described tradition as giving a vote to the most obscure of all classes, our ancestors, and called it the democracy of the dead (Chesterton, 1908). In transplantation the dead do vote, in a literal way. Every deceased donor is a person whose last act was a decision, honored by a family. Any reform that makes that decision feel less safe or less honored has spent capital that cannot easily be earned back. That is the lesson of Chapter 14.

The near. The hospital coordinator who knows the unit. The rural dialysis unit that holds a county in place. The nurse who knows the names of a patient's children. The friend trained to ask on someone's behalf. Wherever I found success in this book, it came from institutions small enough to be recognized. A policy that cannot see them will tend to wear them down.

The household. The kitchen table, the care partner, the home machine. Policy should treat the family as the central unit of care it already is, and stop pretending that unpaid labor is free.

The right to say no. The conservative tradition honors conscience and the individual's authority over her own life. A system that cannot offer, and honestly describe, the choice to forgo dialysis has failed to honor it.

What I would change

A conservative is not a defender of everything. The things I would change are mostly the things nobody can see.

Count the hours. Put the patient's time, travel, recovery, and care partner strain on the dashboard next to the metrics already there. Chapters 1, 8, and 20.

Say it early. Tell people about kidney disease while choices are still wide, and about every option, including transplantation, home therapy, and conservative care. Chapters 5 and 6.

End the receipt. No living donor should pay to give. Chapter 13.

Keep the unit open. Treat the closure of a local dialysis unit as a failure to be investigated and not an outcome to be shrugged at. Chapter 9.

Give the decider a map. Standard information, a travel-time projection, and a second look for the kidney declined again and again. Chapter 15.

Count what is promised. Require independent replication of what manufacturers announce, and report what each new machine gives back in time. Chapter 16.

Be honest about the graft. Tell recipients how long a kidney lasts, plan for the second chapter, and measure whether the drugs are being skipped for cost. Chapters 17 and 19.

Make the family's hallway legible. Explain who determines death, who can halt a donation, and who will call in a month. Chapter 14.

Three tests for any new rule

Chesterton's parable suggests a way of questioning proposals, mine included, before adopting them. I offer three tests, in the form of questions that anyone can ask.

What is the fence? Before changing a practice, can its defenders and its critics both say what it was meant to do? If no one can, the first task is to find out. Many of the rules in transplantation were written to prevent specific harms, and the harms are not always visible to the people who inherit the rules.

Who is nearest? Is the decision being made by the person or the institution closest to those affected, or has it been lifted to a distance without a reason? The coordinator in the Spanish hospital, the friend in the champion program, the nurse in the rural unit, and the family in the hallway are each the nearest competent party. A rule that overrides them should say why.

What does it cost the near? Every reform has a price, and the price often falls on the small institutions that were never consulted: a family that will now have to do the paperwork, a clinic that will lose its late shift, a coordinator who will have a new form to fill. A reform that lowers the cost of the grand design by raising the cost of the household has not saved anything. It has moved the bill.

None of these tests prevents change. They discipline it, which is all that the conservative tradition at its best has ever asked.

What would change my mind

A person who argues for reform should say what would make her retreat. If patients, asked what they want measured, named things I have not imagined, I would change the dashboard to match. If a well-designed trial showed that a technology I have praised gives back no time, I would say so. If a pilot of home dialysis showed that it merely moved burden onto families who could not bear it, I would withdraw

my enthusiasm for it. And if the people who actually sit in the chairs told me that the whole frame of this book, the arithmetic of hours, misses what they need, I would ask them to tell me what it is.

The last conservative argument

Burke's phrase about the little platoon is often taken to mean that the small is good and the large is bad. I think it means something more exact. The large things in life, a nation, a health system, a profession, are made of small things, and they are only as trustworthy as the small things are. The kidney ecosystem is not broken because it lacks a grand design. It is strained because the small things it depends on, the household, the coordinator, the unit, the volunteer, the family in the hallway, have been asked to carry more than they were ever built to carry, with less recognition than they deserve.

The fence, in Chesterton's story, is still standing. The wise reformer goes away and thinks, and when she comes back she can say what the fence is for. What is it for? It is for keeping the people on the road from harm. That is a good thing for a fence to be for. Our task is to make sure it still does.

Epilogue: A Tuesday Worth Wanting

“How vain it is to sit down to write when you have not stood up to live.”

Henry David Thoreau, Journal (1851)

Picture Ruth again, a few years from now, in a story I cannot promise. It is Tuesday. She did not go to the chair this morning, because there is no chair. Perhaps a kidney arrived. Perhaps her treatment moved to her kitchen. Perhaps a friend she trained to ask for her did the asking. I do not know how it happened, and she would tell you the how matters less than the what. It is one o'clock in the afternoon, and she is doing something unimportant. She is deciding whether to make soup.

Nothing in the data in this book predicts that Tuesday for any individual. What the data do show is that the ingredients exist. Medicines that keep the silence from lasting. Machines that fit in a kitchen. Chains of strangers who give. Cells that may teach the body peace. Pig kidneys that have run for months. Programs that train friends. Laws that, belatedly, began to reimburse donors for their

generosity. The ingredients are not yet a meal, and many of them will not survive a clinical trial. But they exist, which was not true a generation ago.

What remains is a decision about what we are building toward. I have argued that the answer is time, and I have tried to say so without sentiment, because sentiment is not what the person in the chair needs. She needs the afternoon back. She needs the care partner counted, the local unit kept open, the cost of the pill known, the choice of whether to dialyze at all offered honestly, and a kidney, when one comes, that lasts.

I have sat in rooms where the chord of the machines hums and no one talks. I have learned that silence is not emptiness. It is attention, spent on one another. The same attention, scaled up, is the only thing that has ever made a system humane. We have a great many instruments in this field. What we need is to be a little more present in the room.

I will leave you with a question, which I ask of myself as often as I ask it of anyone. If you were given a certain number of afternoons, what would you want them for? And when you know the answer, what would you ask of the people who look after you, to help you spend them?

Two things more, and then I will let you go. The first is a thank-you to the people this book is really about, who will never read it, and who are right now sitting in a chair with a book of their own. You taught me what attention is. The second is an invitation. If you are one of those people, or you love one, and you find something in these pages that is

wrong, or incomplete, or that misses the thing you most wanted said, tell the Foundation. A book about being counted ought to be willing to be corrected.

Appendix A: Source Notes

Sources cited in the text as “Source Notes S” and a number are described here, with the grade I assigned. Grade A is a primary document or registry whose key figures I confirmed. Grade B is a credible secondary account. Grade C is a single-source figure or one from a party with a stake in the result. A dagger (†) means that some bibliographic detail, such as a complete author list, should be completed from the original before any future edition. The research tools used for this manuscript returned summaries and bibliographic details and not the full text of every paper, and every figure in the book should be read with that in mind.

Note	Grade	What it supports and where it was found	Still to confirm
S1	A†	Online survey of 359 in-center hemodialysis patients (Kidney Medicine, 2020): median recovery 3 h; 40% at least 4 h; washed-out 62%, cramps 44%, low-BP symptoms 42%.	Title, authors, volume
S2	B	Review statement that typical post-dialysis recovery is 2 to 4 hours and about a quarter report more than 6 hours; French cohort of 536 patients (15% over 12 hours; sodium drop and lower frequency associated with longer recovery).	Primary papers

Note	Grade	What it supports and where it was found	Still to confirm
S3	B†	Cooler dialysate studies: 2024 Pakistani quasi-experimental study (92 patients, 36 vs 37 degrees); older crossover study (50 patients, 35 vs 37 degrees, 76% more energetic); registered Iranian and Turkish trials without published results.	Primary papers
S4	B	Trade report of a prescription-change study: median recovery fell from 210 to 60 minutes at six months and 105 at twelve.	Primary paper
S5	A†	FHN Daily Trial (245 patients): left-ventricular mass reduced by 13.1 g (95% CI 5.0 to 21.3); nocturnal trial not significant but better phosphate and blood pressure control.	NEJM papers
S6	A†	Danish registry study (2005 to 2019) of employment before and after dialysis or transplantation: 11% vs 61% at dialysis start; 20% vs 67% after transplant.	Title, authors
S7	B†	Meta-analysis of employment in dialysis and transplant patients (26.3%; 36.9% before and 38.2% after transplant); US cohort of dialysis starters 1996 to 2013 (23 to 24%).	Primary papers
S8	B†	2025 cross-sectional study of 94 transplant recipients in Poland: 66% resumed work; 41.9% within six months.	Title, authors
S9	B†	Small cross-sectional studies of caregivers of hemodialysis patients from several countries (e.g., Indonesia 2022, Greece, India 2025): hours of care linked to depression and strain.	Pick one or two primary papers to cite

Note	Grade	What it supports and where it was found	Still to confirm
S10	B	Live Donor Champion program (Johns Hopkins): pilot of 15 vs 15 matched candidates; 163-candidate follow-up with 81 referrals and 5.5-fold odds; conference abstracts and registry entry NCT03435510; no randomized trial found.	Published paper(s)
S11	A†	CKDopps analysis (Kidney Medicine, 2023): 1,018 people with CKD, 7% planning conservative management; 18% of 407 surveyed had discussed forgoing dialysis; no provider had a dedicated clinic or protocol.	Title, authors
S12	B†	Qualitative study of 20 Veterans Affairs nephrologists on conservative management.	Title, authors
S13	B†	Editorial and observational literature on conservative management in adults 75 and older (survival and quality of life, hospitalization, willingness to trade life expectancy).	Primary papers
S14	B†	Reviews reporting dialysis withdrawal as a cause of death in 20 to 30% of patients in Western countries.	Primary reviews
S15	B†	Single-center Midwestern cohort 2001 to 2015: 1,226 patients, 536 deaths, 262 after withdrawal; death-form sensitivity about 73%.	Title, authors
S16	B†	Italian program joining nephrology and palliative care (Giornale Italiano di Nefrologia, 2022).	Title, authors
S17	C	Australian conference abstract (ASN Kidney Week, 2023): typical survival about a week after stopping dialysis; most die in acute care.	Full paper if published

Note	Grade	What it supports and where it was found	Still to confirm
S18	C	American Kidney Fund page (updated 2025): 35.5 million Americans; nine in ten unaware. The underlying calculation was not found.	Primary CDC source for the nine-in-ten figure
S19	B†	Study of awareness among adults with CKD stages 3 and 4 (about half unaware).	Title, authors
S20	B†	Analyses of preemptive transplantation: 17% of recipients and 31% of living-donor recipients (2003 to 2012); 2.5% of patients with incident kidney failure in 2015; 2025 UNOS study on benefit.	Primary papers
S21	B	JAMA Internal Medicine analysis of the waiting-time modification policy (more than 181,000 adults; more than 21,000 candidates; 5.3 more transplants per 1,000 listings), via a March 2026 trade report.	Journal article
S22	A/B	Emory patient-navigation randomized trial (401 patients; 32% vs 26% waitlisted; 126 days longer to waitlisting) and RaDIANT dialysis-facility trial (more than 9,000 patients, 134 facilities, Georgia; improved referral, narrowed racial disparity).	Journal papers
S23	B	Peer mentoring: transplant-candidate program (no difference in time to listing or quality of life); 200-patient hemodialysis trial (JASN, April 2025; incidence rate ratio 0.60 vs 0.85, overall not significant); registered feasibility studies.	Journal papers
S24	A†	Registry-based study of home dialysis uptake among incident patients (6.8% in 2010 to 13.3% in 2020).	Title, authors

Note	Grade	What it supports and where it was found	Still to confirm
S25	A†	JCI Insight report of the wearable artificial kidney trial (NCT02280005), stopped after the seventh participant for device problems.	Title, authors
S26	C	A 2026-dated university web page describing the Kidney Project as in animal testing with human trials expected 2026 or 2027; not verified.	Project website or trial registry
S27	B	National Kidney Foundation letter (October 2023): about 170 dialysis facilities closed in a year; median one-way travel increase of 6.6 miles in small-town and rural areas, from older data.	Original letter and cited study
S28	A†	Registry-based study of dialysis facility closures in remote areas, 2001 to 2015: more hospitalizations, missed treatments, marginal mortality increase.	Title, authors
S29	B	Trade-press report of the 2025 nephrology fellowship match (about 73% fill).	Original match data
S30	B†	USRDS-based analysis of rural patients with kidney failure (about 240,000); tele-nephrology case report from a rural critical-access hospital (more than 100 patients in 12 months).	Primary papers
S31	B†	Disaster literature: 2019 systematic review (Americas); Hurricane Katrina survey (16.8% missed 3 or more sessions; 23% hospitalized); Hurricane Maria mortality study.	Primary papers
S32	A†	Cervantes et al., JAMA Internal Medicine (2018): 211 undocumented patients; emergency-only vs standard dialysis.	Confirm authors and article

Note	Grade	What it supports and where it was found	Still to confirm
S33	A†	Nguyen et al., JAMA Internal Medicine (2018): 181 adults; scheduled vs emergency-only dialysis (one-year mortality 3% vs 17%; \$5,768 less per month).	Confirm authors and article
S34	B	Published account of Colorado’s 2019 change to emergency Medicaid for kidney failure (per-patient monthly spending about \$20,000 before, \$5,574 after); totals not reconciled.	Primary state data
S35	B	National Kidney Registry reports of chain no. 124 (2008): 30 transplants, 17 hospitals, 11 states, four months.	Original report
S36	B/C	University of Alabama at Birmingham announcements of its long kidney chain (74 donors and 74 recipients; later 100 or more recipients); current total not confirmed.	Current UAB figure
S37	B	Older analysis of living kidney donation declining after a 2004 peak, most among men, Black adults, and younger and lower-income adults.	Primary analysis
S38	B	2026 article in the Medical Journal of Australia: 253 living-donor kidney transplants in 2024 against 354 in 2008.	Article
S39	A/B	Honor Our Living Donors Act (Consolidated Appropriations Act, 2026, P.L. 119-75, enacted 3 February 2026); Federal Register notices 2026-13250 and 2026-19911; 21% cap figure from a physicians’ group letter.	Text of law, final notice, source of 21%

Note	Grade	What it supports and where it was found	Still to confirm
S40	B	Private living-donor protection programs (wage reimbursement, life insurance) described by the National Kidney Registry and the American Kidney Fund.	Program terms
S41	C	Policy-tracker and campaign figures on the End Kidney Deaths Act (cosponsors, chance of passage).	Congress.gov
S42	B†	Australian interview study of 49 potential donor families; literature review on donor-family grief and satisfaction.	Titles, authors
S43	B†/C	Fifteen-country European inventory of donor-family aftercare; analysis of 1,613 donor-family and recipient letters (conference abstract); Southern California aftercare department abstract.	Primary papers
S44	B	Department of Health and Human Services summary (July 2025): concerns in 103 of 351 cases.	HHS announcement text
S45	B	Literature on offer decisions: median seven offers per kidney; 68-physician study with greater within-center variation; survey with kappa 0.13.	Primary papers
S46	A†	Transplant International, June 2026 (doi 10.3389/ti.2026.15840): 74 kidney pairs from older donors; cold ischemia over 12 h vs under 8 h, odds ratio 6.30 for delayed graft function.	Title, authors

Note	Grade	What it supports and where it was found	Still to confirm
S47	A/B	HRSA organ-transport page; McDermott+ 2024 briefing; UNOS tracking tool (16 OPOs); HRSA monitoring report on the 2021 kidney allocation change (nonuse 21% to 26%); OPTN Kidney Committee summary (28 July 2025, continuous distribution paused).	Current status of each
S48	C	HEMO2life first study (60 grafts, six French centers; 48-month survival 98.3% vs 86%; manufacturer delayed-graft-function figures); registry entries NCT05050513 and NCT04181710.	Journal publication; reconcile registry numbers
S49	A†/B	American Journal of Transplantation (2021) registry analysis of median graft survival; American Transplant Congress abstract with projected half-lives and 5- and 10-year survival.	Titles, authors
S50	A/B	Medeor MDR-101 phase 3 (American Journal of Transplantation, July 2025; ATC 2023 interim): 20 treated, 10 control; about 95% off immunosuppression at one year; 75% for more than two years.	Journal article
S51	A†/B	JAMA Internal Medicine (2013) registry analysis of graft loss by age (ages 14 to 16 highest); English national registry study of young adults.	Titles, authors
S52	A†/B	TransNephro randomized trial (Germany, 2025; 102 participants; no difference in adherence); Stanford single-center transition program abstract.	Titles, authors; graft-survival results

Note	Grade	What it supports and where it was found	Still to confirm
S53	B	Medicare Part B-ID (since December 2022; \$97.10 per month in 2023); 2019 HHS analysis; cost-effectiveness analysis; congressional testimony citing a 2012 report.	Primary documents

What is thin or unfinished

A few matters deserve a plain statement. The ten-year follow-up of the machine-perfusion trial reached me only through the device maker’s announcement. The oxygen-carrier figures are preliminary. The count of living donors in 2025 comes from one news report. The 2026 outcomes of the organ-procurement tiers and the first-year results of the transplant-hospital payment model had not been published. The Medicaid provisions are described by advocacy organizations and a fact brief, and the enacted text should be read directly. The pig-kidney figures after the first formal trial transplant come from companies and the press, apart from the peer-reviewed report of a single patient.

Appendix B: References

Entries marked † have bibliographic details that are partly confirmed and should be completed from the original before any future edition. Works cited in the text only by author and year, such as the epigraph sources, appear at the end.

American Kidney Fund. (2026, May). *Sixth annual living donor protection report card* [News report]. https://www.bakersfield.com/ap/news/american-kidney-funds-sixth-annual-living-donor-protection-report-card-shows-progress/article_2a0c0a4e-f605-59f0-a4f2-eacb06f0bd88.html

American Society of Nephrology. (2025, September). *Kidney News*. https://www.asn-online.org/publications/kidneynews/archives/2025/KN_2025_09_sep.pdf

Artificial kidney breakthrough as bioreactor survives in pig model without immune response. (n.d.). *Renal Interventions*. <https://renalinterventions.net/artificial-kidney-breakthrough-as-bioreactor-survives-in-pig-model-without-immune-response/>

Association of social isolation and loneliness with the incidence and life expectancy of chronic kidney disease: A population-based cohort study. (2026)†. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12825595/>

Associated Press. (2024, October 28). People opt out of organ donation programs after reports of a man mistakenly declared dead. <https://www.bangordailynews.com/2024/10/28/nation/people-opt-out-of-organ-donation-programs-after-reports-of-a-man-mistakenly-declared-dead/>

6 month waiting time modification and 1 year race-neutral eGFR monitoring report. (n.d.). *Organ Procurement and Transplantation Network*. https://hrsa.unos.org/media/k24brtvh/6-month-waiting-time-modification_1-year-race-neutral-egfr-monitoring-report.pdf

- Burke, E. (1790). *Reflections on the revolution in France*. J. Dodsley.
- Centers for Disease Control and Prevention. (2026, March). *Chronic kidney disease in the United States*. <https://www.cdc.gov/kidney-disease/php/data-research>
- Centers for Medicare & Medicaid Services. (n.d.). *Increasing Organ Transplant Access (IOTA) model fact sheet*. <https://www.cms.gov/priorities/innovation/files/iota-model-fs.pdf>
- Centers for Medicare & Medicaid Services. (2025). *ESRD Treatment Choices (ETC) model third evaluation report: Annual evaluation at a glance*. <https://www.cms.gov/priorities/innovation/data-and-reports/2025/etc-3rd-eval-aag-rpt>
- Chesterton, G. K. (1908). *Orthodoxy*. John Lane.
- Chesterton, G. K. (1929). *The thing*. Sheed & Ward.
- ClinicalTrials.gov. (n.d.). *Evaluation of HEMO2Life for kidney graft preservation (OXYOP): 4 years follow-up* (NCT05050513). <https://clinicaltrials.gov/study/NCT05050513>
- CMS releases first-year results of ESRD Treatment Choices model. (2023, August 16). *Healio*. <https://www.healio.com/news/nephrology/20230816/cms-releases-firstyear-results-of-esrd-treatment-choices-model>
- Cold Ischemia Foundation. (n.d.-a). *BHOC: The technology nobody's explaining to patients*. <https://coldischemia.foundation/bhoc.html>
- Cold Ischemia Foundation. (n.d.-b). *The evidence behind BHOC*. <https://coldischemia.foundation/bhoc-evidence.html>
- Debout, A., et al. (2015)†. Each additional hour of cold ischemia time significantly increases the risk of graft failure and mortality following renal transplantation. *Kidney International*, 87(2), 343–349.
- Developing a set of core outcomes for trials in hemodialysis: An international Delphi survey. (2017)†. <https://mdanderson.elsevierpure.com/en/publications/developing-a-set-of-core-outcomes-for-trials-in-hemodialysis-an-i>
- Diabetic kidney disease therapies keep flowing. (n.d.). *The Hospitalist*. <https://blogs.the-hospitalist.org/content/diabetic-kidney-disease-therapies-keep-flowing>

Dialysis care and death following Hurricane Sandy. (n.d.)†. <https://pubmed.ncbi.nlm.nih.gov/25156306/>

Effects of semaglutide with and without concomitant SGLT2 inhibitor use in participants with type 2 diabetes and chronic kidney disease in the FLOW trial. (2024)†. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11485243/>

eGenesis. (2026, September)†. *Patient update on EGEN-2784 expanded-access recipients* [Company press release]. <https://egenesisbio.com/news-media>

Emergence of an Israel faith-based community organization facilitating live donor kidney transplantation. (2018)†. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5992728/>

End Kidney Deaths Act, H.R. 2687, 119th Cong. (2025). <https://legiscan.com/US/drafts/HB2687>

ESRD facility characteristics by rurality and risk of closure. (2017)†. South Carolina Rural Health Research Center. https://caravel.sc.edu/study/colleges_schools/public_health/research/research_centers/sc_rural_health_research_center/documents/161esrdfacilitycharacteristicsbyruralityandriskofclosure.pdf

FDA clears Quanta home hemodialysis system. (n.d.). *MassDevice*. <https://www.massdevice.com/fda-clears-quanta-home-hemodialysis-system/>

Fierce Healthcare. (2025). *HHS moves to decertify Miami organ transplant group amid efforts to reform organ donation system*. <https://www.fiercehealthcare.com/regulatory/hhs-moves-decertify-miami-organ-transplant-group-it-seeks-reform-organ-donation-system>

GigaFact. (2026, August 25). *Fact brief on Medicaid work requirements and eligibility redeterminations*†. <https://gigaFact.org/?p=18047>

Health Resources and Services Administration. (n.d.). *Organ transport*. <https://www.hrsa.gov/optn/professionals/resources/organ-transport>

Health Resources and Services Administration. (2026a, July 1)†. *Notice requesting comment on living organ donor reimbursement* (Federal Register public inspection document 2026-13250). <https://public-inspection.federalregister.gov/2026-13250.pdf>

- Health Resources and Services Administration. (2026b, August). *Building a safer, more accountable transplant system*. <https://www.hrsa.gov/optn-modernization/updates/august-2026>
- Health Resources and Services Administration. (2026c, May). *OPTN board updates: New director, extensions for president and vice president, membership status*. <https://www.hrsa.gov/optn/news-events/news/may-2026-board-of-director-updates>
- Health Resources and Services Administration. (2026d, September). *Advancing a safer, more transparent organ donation, procurement, and transplantation system*. <https://www.hrsa.gov/optn-modernization/updates/september-2026>
- Health Resources and Services Administration, Division of Transplantation. (2025, March 24). *Information memo to the Associate Administrator: Kentucky Organ Donor Affiliates*. https://d1dth6e84htgma.cloudfront.net/Attachment_1_Information_Memo_to_the_Associate_Administrator_KYDA_8939df6443.pdf
- Holland & Knight LLP. (2026, March). *CMS issues additional guidance on the organ donation process*. <https://www.hklaw.com/en/insights/publications/2026/03/cms-issues-additional-guidance-on-the-organ-donation-process>
- Hosgood, S. A., et al. (2023)†. Normothermic machine perfusion versus static cold storage in donation after circulatory death kidney transplantation: A randomized controlled trial. *Nature Medicine*, 29, 1511–1519. <https://doi.org/10.1038/s41591-023-02376-7>
- Immune profiling in a living human recipient of a gene-edited pig kidney. (2025). *Nature Medicine*. <https://www.nature.com/articles/s41591-025-04053-3>
- Kidney disease affects 1 in 7 Americans, says CDC. (n.d.). *Becker's Hospital Review*. <https://www.beckershospitalreview.com/quality/kidney-disease-affects-1-in-7-americans-says-cdc/>
- La Moncloa. (2025, January 16). *Spain, with over 6,400 transplants, exceeds forecasts for 2024*. <https://www.lamoncloa.gob.es/lang/en/gobierno/news/paginas/2025/20250116-transplant-figures.aspx>

- Matthews. (2025). *Dialysis centers: 2025 update* [Market insight].
https://www.matthews.com/market_insights/dialysis-centers-2025-update
- Ministerio de Sanidad. (2026, January). *España vuelve a superar los 6.300 trasplantes en 2025*. <https://www.sanidad.gob.es/va/gabinete/notasPrensa.do?id=6828>
- Moers, C., et al. (2009). Machine perfusion or cold storage in deceased-donor kidney transplantation. *New England Journal of Medicine*, 360(1), 7–19.
<https://doi.org/10.1056/NEJMoa0802289>
- Nation's longest single-site kidney chain reaches 100. (n.d.). *Newswise*.
<https://www.newswise.com/articles/nation's-longest-single-site-kidney-chain-reaches-100>
- National Kidney Foundation. (2025, July). *Big Beautiful Bill is downright ugly for kidney patients*. <https://www.kidney.org/press-room/big-beautiful-bill-downright-ugly-kidney-patients>
- National Kidney Registry. (n.d.). *National Kidney Registry facilitates 10,000th living donor kidney transplant*. <https://kidneyregistry.com/news/national-kidney-registry-facilitates-10000th-living-donor-kidney-transplant>
- National Organ Transplant Act, Pub. L. No. 98-507, 98 Stat. 2339 (1984).
- Nature Reviews Nephrology*. (2025)†. Review of kidney preservation methods and outcomes. <https://doi.org/10.1038/s41581-025-00993-8>
- Normothermic versus hypothermic machine perfusion in kidney transplantation: A randomized controlled trial. (2026)†. *American Journal of Transplantation*.
[https://www.amjtransplant.org/article/S1600-6135\(25\)03191-0/fulltext](https://www.amjtransplant.org/article/S1600-6135(25)03191-0/fulltext)
- Older patients and conservative kidney management. (2026)†. *JAMA Network Open*.
<https://doi.org/10.1001/jamanetworkopen.2026.36332>
- Organ Procurement and Transplantation Network. (n.d.). *OPTN Board approves waiting time adjustment for kidney transplant candidates affected by race-based calculation*. <https://optn.transplant.hrsa.gov/news/optn-board-approves-waiting-time-adjustment-for-kidney-transplant-candidates-affected-by-race-based-calculation>

Organ Recovery Systems. (2025, November 5). *Organ Recovery Systems announces publication of 10-year follow-up data of landmark hypothermic machine perfusion study featuring LifePort Kidney Transporter in the New England Journal of Medicine* [Press release]. Business Wire.

<https://www.businesswire.com/news/home/20251105409077/en>

Patient who received pig kidney becomes first to progress to human kidney transplant. (2026). *Harvard Medical School News*.
<https://hms.harvard.edu/news/patient-who-received-pig-kidney-becomes-first-progress-human-kidney-transplant>

PhillyVoice. (2026)†. *Article on living organ donors and 2025 registry figures*.
<https://www.phillyvoice.com/daryl-hall-kidney-transplant-living-organ-donors/>

Prevalence of depression in hemodialysis patients: A meta-analysis. (2025)†. *Seminars in Dialysis*. <https://doi.org/10.1111/sdi.70043>

Quantifying center-level aggressiveness in transplanting suboptimal, deceased donor kidneys in the United States. (n.d.)†.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12795041/>

Rees, M. A., et al. (2009)†. A nonsimultaneous, extended, altruistic-donor chain. *New England Journal of Medicine*, 360(11), 1096–1101.

Scientific Registry of Transplant Recipients, & Organ Procurement and Transplantation Network. (2025). *OPTN/SRTR 2023 annual data report: Introduction*. <https://srtr.hrsa.gov/adr/2023/Introduction>

Scientific Registry of Transplant Recipients, & Organ Procurement and Transplantation Network. (2026a). *OPTN/SRTR 2024 annual data report: Overview of US solid organ transplantation*. <https://srtr.hrsa.gov/adr/2024/Overview/>

Scientific Registry of Transplant Recipients, & Organ Procurement and Transplantation Network. (2026b). *OPTN/SRTR 2024 annual data report: Kidney. American Journal of Transplantation*. [https://www.amjtransplant.org/article/S1600-6135\(26\)00985-8/fulltext](https://www.amjtransplant.org/article/S1600-6135(26)00985-8/fulltext)

Strathern, M. (1997). “Improving ratings”: Audit in the British university system. *European Review*, 5(3), 305–321.

- The pediatric nephrology workforce is in crisis. (2025). *Healio*.
<https://cm.healio.com/news/nephrology/20250207/nni0325weidemannfirstworldfinal>
- The rural nephrology workforce shortage is already here. (2026, September).
National Rural Health Association. <https://www.ruralhealth.us/blogs/2026/09-september/the-rural-nephrology-workforce-shortage-is-already-here>
- The Spanish model. (2025). *Healthcare in Europe*. <https://healthcare-in-europe.com/en/news/the-spanish-model.html>
- They counted on a rural dialysis unit to keep them alive. Then it closed. (2026, April 15). *KFF Health News*. <https://www.vpm.org/2026-04-15/they-counted-on-a-rural-dialysis-unit-to-keep-them-alive-then-it-closed>
- Transplant tolerance: The frontier of kidney care. (2026, April 23)†. *HCPLive*.
<https://www.hcplive.com/view/transplant-tolerance-frontier-kidney-care-joseph-leventhal>
- Tushla, L., et al. (2015)†. Living-donor kidney transplantation: Reducing financial barriers to live kidney donation. *Clinical Journal of the American Society of Nephrology*, 10(9), 1696–1702.
- United States Renal Data System. (2014). *2014 annual data report, chapter 5: Mortality*. National Institute of Diabetes and Digestive and Kidney Diseases.
https://www.niddk.nih.gov/-/media/Files/USRDS/2014-Previous-ADR/Volume-2/Chapters/v2_ch_05_mortality_14.pdf
- United Therapeutics Corporation. (2025, November 3). *United Therapeutics Corporation announces first transplant in EXPAND clinical trial of UKidney in patients with end stage renal disease* [Press release]. Business Wire.
<https://www.businesswire.com/news/home/20251103435431/en>
- What nephrologists learned from Hurricane Sandy. (2015, July 9)†. *AJKD Blog*.
<https://ajkdblog.org/2015/07/09/what-nephrologists-learned-from-hurricane-sandy>
- Wolfe, R. A., et al. (1999). Comparison of mortality in all patients on dialysis, patients on dialysis awaiting transplantation, and recipients of a first cadaveric transplant. *New England Journal of Medicine*, 341(23), 1725–1730.

Xenotransplantation in living human recipients: Where we stand in 2026. (2026).
American Journal of Transplantation. [https://www.amjtransplant.org/article/S1600-6135\(26\)00130-9/fulltext](https://www.amjtransplant.org/article/S1600-6135(26)00130-9/fulltext)

Sources of the epigraphs

All epigraphs are quotations from works in the public domain. Pascal, B. (1908 translation by W. F. Trotter). *Pensées*, no. 139. The Holy Bible, King James Version (1611): Psalms 90:12 and 127:2; Ecclesiastes 1:18, 3:1, 3:22, 7:2 and 11:9; Proverbs 22:3 and 22:7. Eliot, G. (1872). *Middlemarch*, Finale. Florio, J. (1603 translation). Montaigne, M. de, *Essays*, I.20. Frost, R. (1914). The death of the hired man, in *North of Boston*. Cather, W. (1913). *Opioneers!* Dante Alighieri (1867 translation by H. W. Longfellow). *Inferno*, Canto I. Emerson, R. W. (1844). Gifts, in *Essays: Second series*. Thoreau, H. D. (1854). *Walden*; (1851). *Journal*. James, W. (1890). *The principles of psychology*. Johnson, S. (1750). *The Rambler*, no. 2. Peabody, F. W. (1927). The care of the patient. *JAMA*, 88(12), 877–882. Chesterton, G. K. (1929). *The thing*. Hippocratic Oath (1923 translation by W. H. S. Jones). Virgil (29 BCE). *Georgics*, II.490.

About the Author

Jeff Parke is the founder of the Cold Ischemia Foundation, an independent advocacy organization based in Ellenton, Florida, for patients, living donors, and care partners in the kidney and transplant communities. The Foundation accepts no donations, grants, or sponsorships and charges posted fees only for services that individuals request. The author has spent the better part of two decades inside the transplant system, in more than one role, and writes from the conviction that the people in these rooms deserve to be counted.